



Welcome to A Comprehensive Course on the Wave Principle

Market prices move in recognizable patterns.

Recent scientific research has validated several aspects of R.N. Elliott's observations about markets. Of course, these findings only confirmed what countless traders and investors knew all along.

R.N. discovered the Wave Principle nearly 70 years ago. Over the decades, his discovery was kept alive by a handful of individuals. A few of those, such as Bolton, Prechter and Frost, educated investors on how to use pattern analysis in financial markets.

This is your opportunity to learn what they know and use what they've used.

What You Will Learn

Successful market timing depends upon learning the patterns of crowd behavior. By anticipating the crowd, you can avoid becoming a part of it.

Our Elliott wave tutorial describes these patterns and explains how they relate to one another. Over the years, we have taught **thousands** of people how to use this method to analyze the markets they follow.

To save you time, our tutorial distills decades of experience in making Elliott waves discernable. We've included everything you need to learn it and use it.

The course is comprehensive, with the same content you'd receive in a formal training class. The big difference is that you can set your own pace and review the material as many times as you like.

Some people tell us that the value of this information to their investment strategy is immeasurable and ask us not to let the secret out. To be frank, there are others who think it's nonsense. Well, that's why we offer it to you **free**. You can make up your own mind. We aren't the least bit worried about "everyone" using Elliott because people are people, and only one in a hundred has the mental toughness to believe his own work.

Ready to get started? Simply navigate to [Lesson 1](#) and begin.

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Lesson 1

1.1 Introduction: The Broad Concept

In *The Elliott Wave Principle – A Critical Appraisal*, Hamilton Bolton made this opening statement

As we have advanced through some of the most unpredictable economic climate imaginable, covering depression, major war, and postwar reconstruction and boom, I have noted how well Elliott's Wave Principle has fitted into the facts of life as they have developed, and have accordingly gained more confidence that this Principle has a good quotient of basic value.

"The Wave Principle" is Ralph Nelson Elliott's discovery that social, or crowd, behavior trends and reverses in recognizable patterns. Using stock market data as his main research tool, Elliott discovered that the ever-changing path of stock market prices reveals a structural design that in turn reflects a basic harmony found in nature. From this discovery, he developed a rational system of market analysis. Elliott isolated thirteen patterns of movement, or "waves," that recur in market price data and are repetitive in form, but are not necessarily repetitive in time or amplitude. He named, defined and illustrated the patterns. He then described how these structures link together to form larger versions of those same patterns, how they in turn link to form identical patterns of the next larger size, and so on. In a nutshell, then, the Wave Principle is a catalog of price patterns and an explanation of where these forms are likely to occur in the overall path of market development. Elliott's descriptions constitute a set of empirically derived rules and guidelines for interpreting market action. Elliott claimed predictive value for The Wave Principle, which now bears the name, "The Elliott Wave Principle."

1.2 Short History

Although it is the best forecasting tool in existence, the Wave Principle is not primarily a forecasting tool; it is a detailed description of how markets behave. Nevertheless, that description does impart an immense amount of knowledge about the market's position within the behavioral continuum and therefore about its probable ensuing path. The primary value of the Wave Principle is that it provides a context for market analysis. This context provides both a basis for disciplined thinking and a perspective on the market's general position and outlook. At times, its accuracy in identifying, and even anticipating, changes in direction is almost unbelievable. Many areas of mass human activity follow the Wave Principle, but the stock market is where it is most popularly applied. Indeed, the stock market considered alone is far more important than it seems to casual observers. The level of aggregate stock prices is a direct and immediate measure of the popular valuation of man's total productive capability. That this valuation has form is a fact of profound implications that will ultimately revolutionize the social sciences. That, however, is a discussion for another time.

R.N. Elliott's genius consisted of a wonderfully disciplined mental process, suited to studying charts of the Dow Jones Industrial Average and its predecessors with such thoroughness and precision that he could construct a network of principles that covered all market action known to him up to the mid-1940s. At that time, with the Dow in the 100s, Elliott predicted a great bull market for the next several decades that would exceed all expectations at a time when most investors felt it impossible that the Dow could even better its 1929 peak. As we shall see, phenomenal stock market forecasts, some of pinpoint accuracy years in advance, have accompanied the history of the application of the Elliott Wave approach.

Elliott had theories regarding the origin and meaning of the patterns he discovered, which we will present and expand upon in Lessons 16-19. Until then, suffice it to say that the patterns described in Lessons 1-15 have stood the test of time.

Often one will hear several different interpretations of the market's Elliott Wave status, especially when cursory, off-the-cuff studies of the averages are made by latter day experts.

However, most uncertainties can be avoided by keeping charts on both arithmetic and semilogarithmic scale and by taking care to follow the rules and guidelines as laid down in this course. Welcome to the world of Elliott.

1.3 Basic Tenets

Under the Wave Principle, every market decision is both produced by meaningful information and produces meaningful information. Each transaction, while at once an effect, enters the fabric of the market and, by communicating transactional data to investors, joins the chain of causes of others' behavior. This feedback loop is governed by man's social nature, and since he has such a nature, the process generates forms. As the forms are repetitive, they have predictive value.

Sometimes the market appears to reflect outside conditions and events, but at other times it is entirely detached from what most people assume are causal conditions. The reason is that the market has a law of its own. It is not propelled by the linear causality to which one becomes accustomed in the everyday experiences of life. Nor is the market the cyclically rhythmic machine that some declare it to be. Nevertheless, its movement reflects a structured formal progression.

That progression unfolds in waves. Waves are patterns of directional movement. More specifically, a wave is any one of the patterns that naturally occur under the Wave Principle, as described in Lessons 1-9 of this course.

The Five Wave Pattern

In markets, progress ultimately takes the form of five waves of a specific structure. Three of these waves, which are labeled 1, 3 and 5, actually effect the directional movement. They are separated by two countertrend interruptions, which are labeled 2 and 4, as shown in Figure 1-1. The two interruptions are apparently a requisite for overall directional movement to occur.

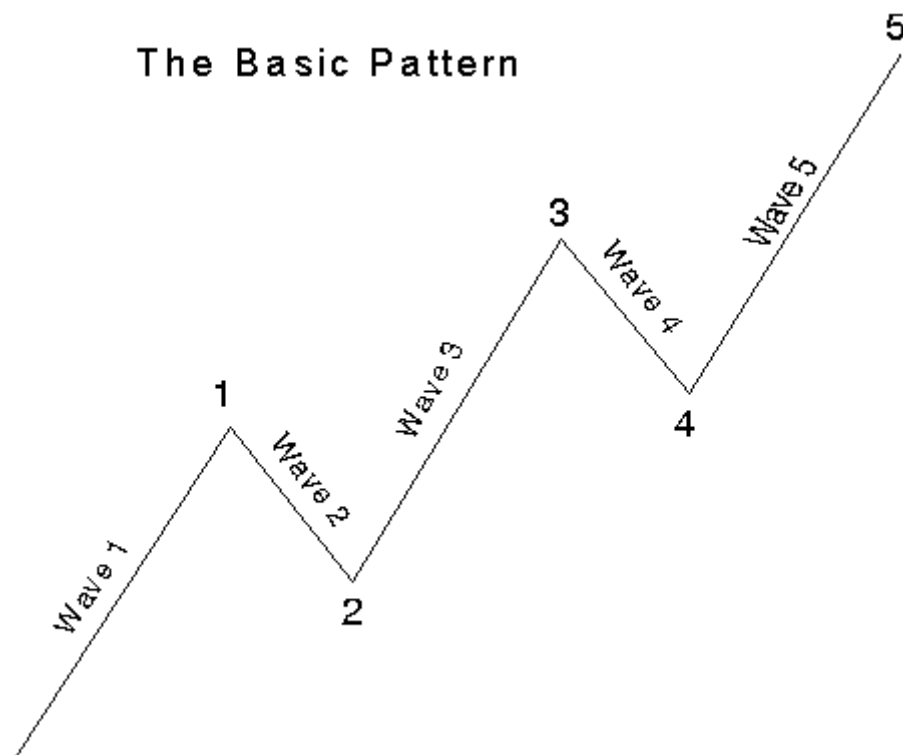


Figure 1-1

R.N. Elliott did not specifically state that there is only one overriding form, the "five wave" pattern, but that is undeniably the case. At any time, the market may be identified as being somewhere in the basic five wave pattern at the largest degree of trend. Because the five wave pattern is the overriding form of market progress, all other patterns are subsumed by it.

1.4 Wave Mode

There are two modes of wave development: motive and corrective. Motive waves have a five wave structure, while corrective waves have a three wave structure or a variation thereof. Motive mode is employed by both the five wave pattern of Figure 1-1 and its same-directional components, i.e., waves 1, 3 and 5. Their structures are called "motive" because they powerfully impel the market. Corrective mode is employed by all countertrend interruptions, which include waves 2 and 4 in Figure 1-1. Their structures are called "corrective" because they can accomplish only a partial retracement, or "correction," of the progress achieved by any preceding motive wave. Thus, the two modes are fundamentally different, both in their roles and in their construction, as will be detailed throughout this course.

In his 1938 book, *The Wave Principle*, and again in a series of articles published in 1939 by *Financial World* magazine, R.N. Elliott pointed out that the stock market unfolds according to a basic rhythm or pattern of five waves up and three waves down to form a complete cycle of eight waves. The pattern of five waves up followed by three waves down is depicted in Figure 1-2.

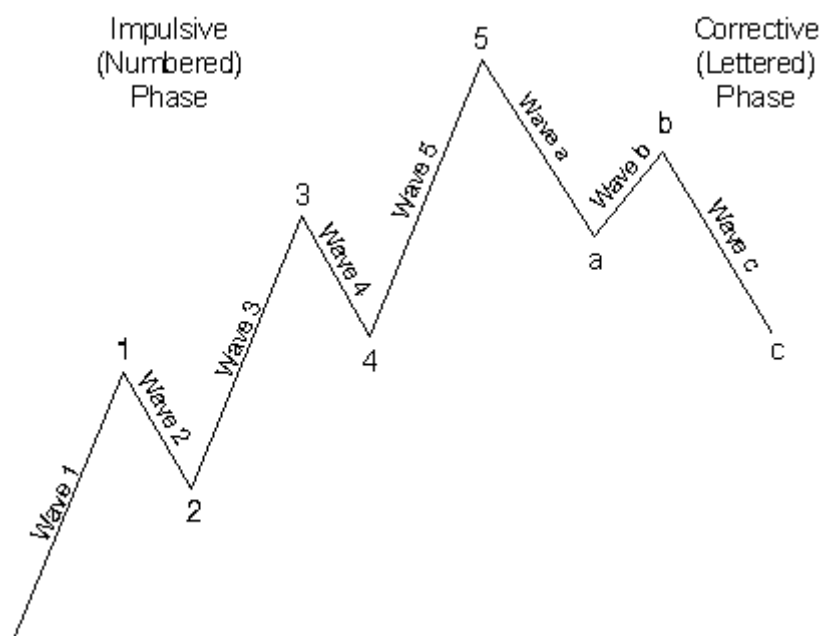


Figure 1-2

One complete cycle consisting of eight waves, then, is made up of two distinct phases, the motive phase (also called a "five"), whose subwaves are denoted by numbers, and the corrective phase (also called a "three"), whose subwaves are denoted by letters. The sequence a, b, c corrects the sequence 1, 2, 3, 4, 5 in Figure 1-2.

At the terminus of the eight-wave cycle shown in Figure 1-2 begins a second similar cycle of five upward waves followed by three downward waves. A third advance then develops, also consisting of five waves up. This third advance completes a five wave movement of one degree larger than the waves of which it is composed. The result is as shown in Figure 1-3 up to the peak labeled (5).

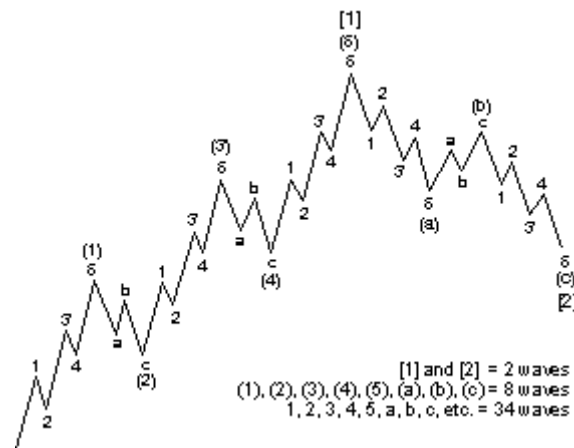


Figure 1-3

At the peak of wave (5) begins a down movement of correspondingly larger degree, composed once again of three waves. These three larger waves down "correct" the entire movement of five larger waves up. The result is another complete, yet larger, cycle, as shown in Figure 1-3. As Figure 1-3 illustrates, then, *each same-direction component of a motive wave, and each full-cycle component* (i.e., waves 1 + 2, or waves 3 + 4) *of a cycle, is a smaller version of itself.*

It is crucial to understand an essential point: Figure 1-3 not only illustrates a *larger* version of Figure 1-2, it also illustrates *Figure 1-2 itself*, in greater detail. In Figure 1-2, each subwave 1, 3 and 5 is a motive wave that will subdivide into a "five," and

each subwave 2 and 4 is a corrective wave that will subdivide into an a, b, c. Waves (1) and (2) in Figure 1-3, if examined under a "microscope," would take the same form as waves [1]* and [2]. All these figures illustrate the phenomenon of constant form within ever-changing degree.

1.5 Essential Design

The market's compound construction is such that two waves of a particular degree subdivide into eight waves of the *next* lower degree, and those eight waves subdivide in exactly the same manner into thirty-four waves of the next lower degree. The Wave Principle, then, reflects the fact that waves of any degree in any series always subdivide and re-subdivide into waves of lesser degree and simultaneously are components of waves of higher degree. Thus, we can use Figure 1-3 to illustrate two waves, eight waves or thirty-four waves, depending upon the degree to which we are referring.

Now observe that within the corrective pattern illustrated as wave [2] in Figure 1-3, waves (a) and (c), which point downward, are composed of five waves: 1, 2, 3, 4 and 5. Similarly, wave (b), which points upward, is composed of three waves: a, b and c. This construction discloses a crucial point: that motive waves do not always point upward, and corrective waves do not always point downward. The mode of a wave is determined not by its absolute direction but primarily by its *relative* direction. Aside from four specific exceptions, which will be discussed later in this course, waves divide in *motive* mode (five waves) when trending in the *same* direction as the wave of one larger degree of which it is a part, and in *corrective* mode (three waves or a variation) when trending in the opposite direction. Waves (a) and (c) are motive, trending in the *same* direction as wave [2]. Wave (b) is corrective because it corrects wave (a) and is *countertrend* to wave [2]. In summary, the essential underlying tendency of the Wave Principle is that action in the same direction as the one larger trend develops in five waves, while reaction against the one larger trend develops in three waves, *at all degrees of trend.*

*Note: For this course, all Primary degree numbers and letters normally denoted by circles are shown with brackets.

Essential Concepts

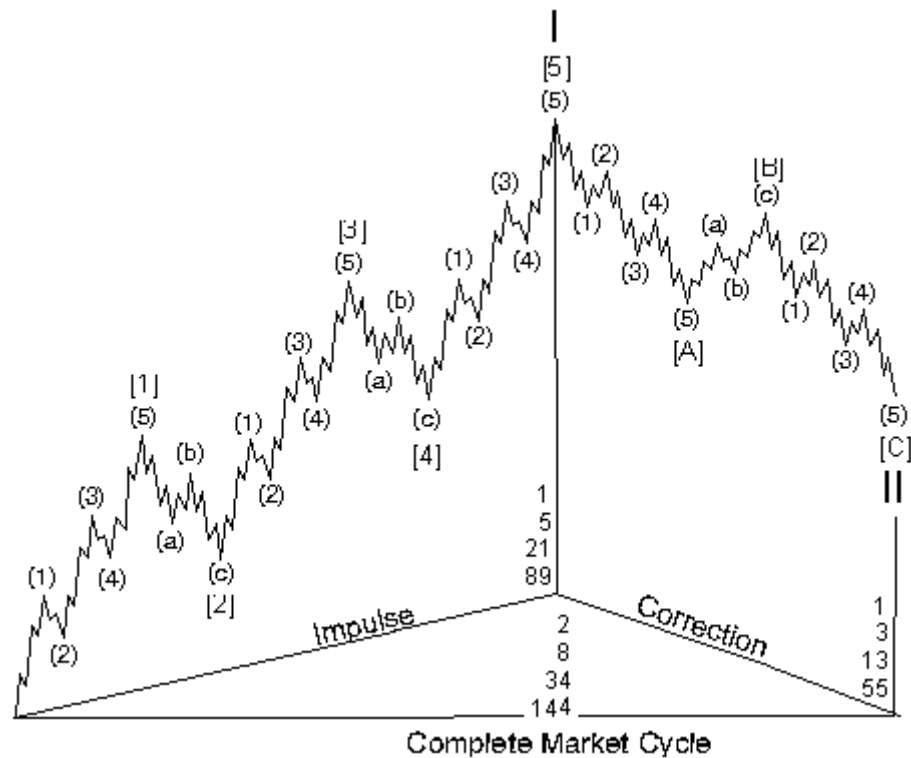


Figure 1-4

The phenomena of *form*, *degree* and *relative direction* are carried one step further in Figure 1-4. This illustration reflects the general principle that in any market cycle, waves will subdivide as shown in the following table.

1.6 Wave Numbers

Number of Waves at Each Degree
Impulse + Correction = Cycle

Largest waves $1+1=2$

Largest subdivisions $5+3=8$

Next subdivisions $21+13=34$

Next subdivisions $89+55=144$

As with Figures 1-2 and 1-3 in Lesson 2, neither does Figure 1-4 imply finality. As before, the termination of yet another eight wave movement (five up and three down) completes a cycle that automatically becomes two subdivisions of the wave of *next* higher degree. As long as progress continues, the process of building to greater degrees continues. The reverse process of subdividing into lesser degrees apparently continues indefinitely as well. As far as we can determine, then, all waves both *have* and *are* component waves.

Elliott himself never speculated on why the market's essential form was five waves to progress and three waves to regress. He simply noted that that was what was happening. Does the essential form

have to be five waves and three waves? Think about it and you will realize that this is *the* minimum requirement for, and therefore the most efficient method of, achieving both *fluctuation* and *progress* in linear movement. *One* wave does not allow *fluctuation*. The fewest subdivisions to create fluctuation is three waves. Three waves *in both directions* does not allow *progress*. To progress in one direction despite periods of regress, movements in the main trend must be at least five waves, simply to cover more ground than the three waves and still contain fluctuation. While there could be more waves than that, the most efficient form of punctuated progress is 5-3, and nature typically follows the most efficient path.

Variations on the Basic Theme

The Wave Principle would be simple to apply if the basic theme described above were the complete description of market behavior. However, the real world, fortunately or unfortunately, is not so simple. From here through Lesson 15, we will fill out the description of how the market behaves in reality. That's what Elliott set out to describe, and he succeeded in doing so.

WAVE DEGREE

All waves may be categorized by relative size, or degree. Elliott discerned nine degrees of waves, from the smallest wiggle on an hourly chart to the largest wave he could assume existed from the data then available. He chose the names listed below to label these degrees, from largest to smallest:

Grand Supercycle
Supercycle
Cycle
Primary
Intermediate
Minor
Minute
Minuette
Subminuette

It is important to understand that these labels refer to specifically identifiable degrees of waves. For instance, when we refer to the U.S. stock market's rise from 1932, we speak of it as a Supercycle with subdivisions as follows:

1932-1937 the first wave of Cycle degree

1937-1942 the second wave of Cycle degree

1942-1966 the third wave of Cycle degree

1966-1974 the fourth wave of Cycle degree

1974-19?? the fifth wave of Cycle degree

Cycle waves subdivide into Primary waves that subdivide into Intermediate waves that in turn subdivide into Minor and sub-Minor waves. By using this nomenclature, the analyst can identify precisely the position of a wave in the overall progression of the market, much as longitude and latitude are used to identify a geographical location. To say, "the Dow Jones Industrial Average is in Minute wave v of Minor wave 1 of Intermediate wave (3) of Primary wave [5] of Cycle wave I of Supercycle wave (V) of the current Grand Supercycle" is to identify a specific point along the progression of market history.

1.7 Degrees

When numbering and lettering waves, the scheme shown below is recommended to differentiate the degrees of waves in the stock market's progression:



The most desirable form for a scientist is usually something like $1_1, 1_2, 1_3, 1_4, 1_5$, etc., with subscripts denoting degree, but it's a nightmare to read such notations on a chart. The above table provides for rapid visual orientation. Charts may also use color as an effective device for differentiating degree.

In Elliott's suggested terminology, the term "Cycle" is used as a name denoting a specific degree of wave and is not intended to imply a cycle in the typical sense. The same is true of the term "Primary," which in the past has been used loosely by Dow Theorists in phrases such as "primary swing" or "primary bull market." The specific terminology is not critical to the identification of relative degrees, and the authors have no argument with amending the terms, although out of habit we have become comfortable with Elliott's nomenclature.

The precise identification of wave degree in "current time" application is occasionally one of the difficult aspects of the Wave Principle. Particularly at the start of a new wave, it can be difficult to decide what degree the initial smaller subdivisions are. The main reason for the difficulty is that wave degree is not based upon specific price or time lengths. Waves are dependent upon *form*, which is a function of both price *and* time. The degree of a form is determined by its size and position *relative to component, adjacent and encompassing waves*.

This relativity is one of the aspects of the Wave Principle that make real time interpretation an intellectual challenge. Fortunately, the precise degree is usually irrelevant to successful forecasting since it is *relative* degree that matters most. Another challenging aspect of the Wave Principle is the variability of forms, as described through Lesson 9 of this course.

1.8 Wave Function

Every wave serves one of two functions: *action* or *reaction*. Specifically, a wave may either advance the cause of the wave of one larger degree or interrupt it. The function of a wave is determined by its *relative direction*. An *actionary* or *trend* wave is any wave that trends in the *same* direction as the wave of one larger degree of which it is a part. A *reactionary* or *countertrend* wave is any wave that trends in the direction *opposite* to that of the wave of one larger degree of which it is part. Actionary waves are labeled with *odd* numbers and letters. Reactionary waves are labeled with even numbers and letters.

All reactionary waves develop in corrective mode. If all actionary waves developed in motive mode, then there would be no need for different terms. Indeed, most actionary waves do subdivide into five waves. However, as the following sections reveal, a few actionary waves develop in corrective mode, i.e., they subdivide into *three* waves or a variation thereof. A detailed knowledge of pattern construction is required before one can draw the distinction between *actionary* function and *motive* mode, which in the underlying model introduced so far are indistinct. A thorough understanding of the forms detailed in the next five lessons will clarify why we have introduced these terms to the Elliott Wave lexicon.

Lesson 2

2.1 Motive Waves

Motive waves subdivide into *five* waves with certain characteristics and always move in the same direction as the trend of one larger degree. They are straightforward and relatively easy to recognize and interpret.

Within motive waves, wave 2 never retraces more than 100% of wave 1, and wave 4 never retraces more than 100% of wave 3. Wave 3, moreover, always travels beyond the end of wave 1. The goal of a motive wave is to make progress, and these rules of formation assure that it will.

Elliott further discovered that in *price* terms, wave 3 is often the longest and never the shortest among the three actionary waves (1, 3 and 5) of a motive wave. As long as wave 3 undergoes a greater percentage movement than either wave 1 or 5, this rule is satisfied. It almost always holds on an arithmetic basis as well. There are two types of motive waves: *impulses* and *diagonal triangles*.

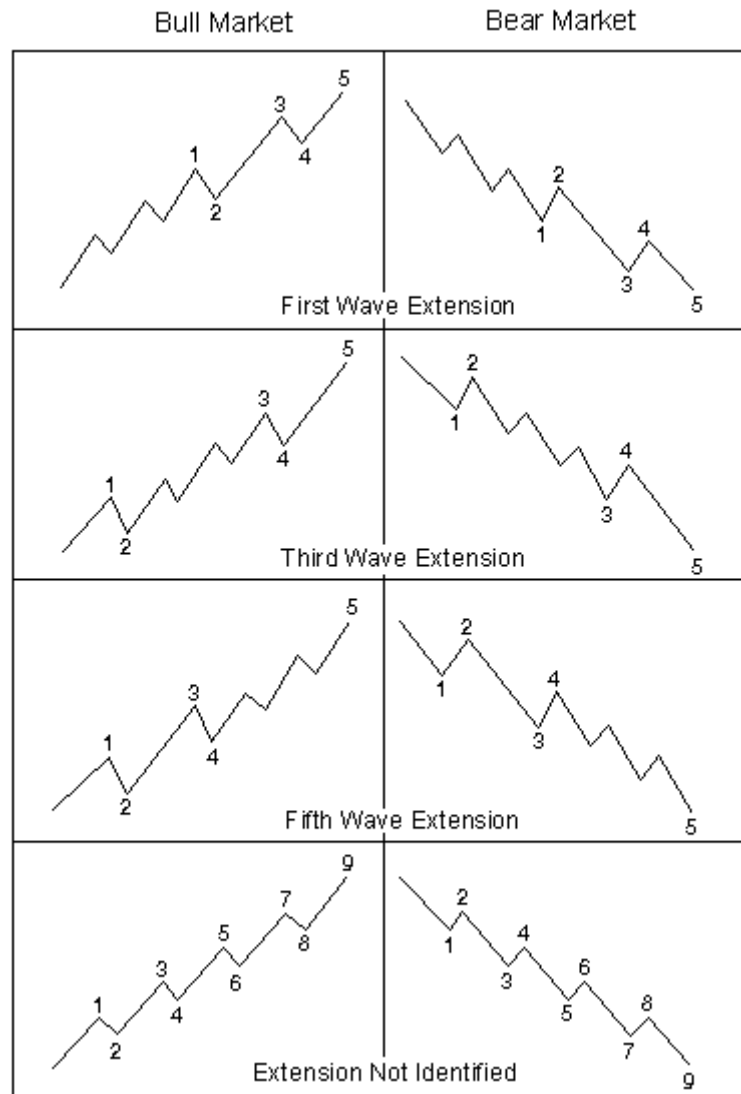
Impulse

The most common motive wave is an *impulse*. In an impulse, wave 4 does not enter the territory of (i.e., "overlap") wave 1. This rule holds for all non-leveraged "cash" markets. Futures markets, with their extreme leverage, can induce short term price extremes that would not occur in cash markets. Even so, overlapping is usually confined to daily and intraday price fluctuations and even then is extremely rare. In addition, the actionary subwaves (1, 3 and 5) of an impulse are themselves motive, and subwave 3 is specifically an impulse. Figures 1-2 and 1-3 in Lesson 2 and 1-4 in Lesson 3 all depict impulses in the 1, 3, 5, A and C wave positions.

As detailed in the preceding three paragraphs, there are only a few simple rules for interpreting impulses properly. A *rule* is so called because it governs all waves to which it applies. Typical, yet *not inevitable*, characteristics of waves are called *guidelines*. Guidelines of impulse formation, including extension, truncation, alternation, equality, channeling, personality and ratio relationships are discussed below and through Lesson 24 of this course. A rule should never be disregarded. In many years of practice with countless patterns, the authors have found but one instance above Subminuette degree when all other rules and guidelines combined to suggest that a rule was broken. Analysts who routinely break any of the rules detailed in this section are practicing some form of analysis other than that guided by the Wave Principle. These rules have great practical utility in correct counting, which we will explore further in discussing extensions.

2.2 Extension

Most impulses contain what Elliott called an extension. Extensions are elongated impulses with exaggerated subdivisions. The vast majority of impulse waves do contain an extension in one and only one of their three actionary subwaves. At times, the subdivisions of an extended wave are nearly the same amplitude and duration as the other four waves of the larger impulse, giving a total count of nine waves of similar size rather than the normal count of "five" for the sequence. In a nine-wave sequence, it is occasionally difficult to say which wave extended. However, it is usually irrelevant anyway, since under the Elliott system, a count of nine and a count of five have the same technical significance. The diagrams in Figure 1-5, illustrating extensions, will clarify this point.



The fact that extensions typically occur in only one actionary subwave provides a useful guide to the expected lengths of upcoming waves. For instance, if the first and third waves are of about equal length, the fifth wave will likely be a protracted surge. (In waves below Primary degree, a developing fifth wave extension will be confirmed by new high volume, as described in Lesson 13 under "Volume.") Conversely, if wave three extends, the fifth should be simply constructed and resemble wave one.

In the stock market, *the most commonly extended wave is wave 3*. This fact is of particular importance to real time wave interpretation when considered in conjunction with two of the rules of impulse waves: that wave 3 is never the shortest actionary wave, and that wave 4 may not overlap wave 1. To clarify, let us assume two situations involving an improper middle wave, as illustrated in Figures 1-6 and 1-7.

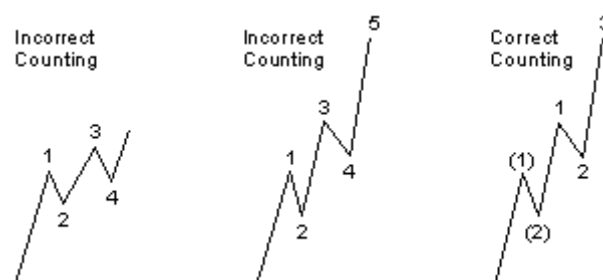


Figure 1-6

Figure 1-7

Figure 1-8

In Figure 1-6, wave 4 overlaps the top of wave 1. In Figure 1-7, wave 3 is shorter than wave 1 and shorter than wave 5. According to the rules, neither is an acceptable labeling. Once the apparent wave 3 is proved unacceptable, it must be relabeled in some way that is acceptable. In fact, it is almost always to be labeled as shown in Figure 1-8, implying an extended wave (3) in the making. Do not hesitate to get into the habit of labeling the early stages of a third wave extension. The exercise will prove highly rewarding, as you will understand from the discussion under Wave Personality in Lesson 14. Figure 1-8 is perhaps the single most useful guide to real time impulse wave counting in this course.

Extensions may also occur within extensions. In the stock market, the third wave of an extended third wave is typically an extension as well, producing a profile such as shown in Figure 1-9. Figure 1-10 illustrates a fifth wave extension of a fifth wave extension. Extended fifths are fairly uncommon except in bull markets in commodities covered in Lesson 28.

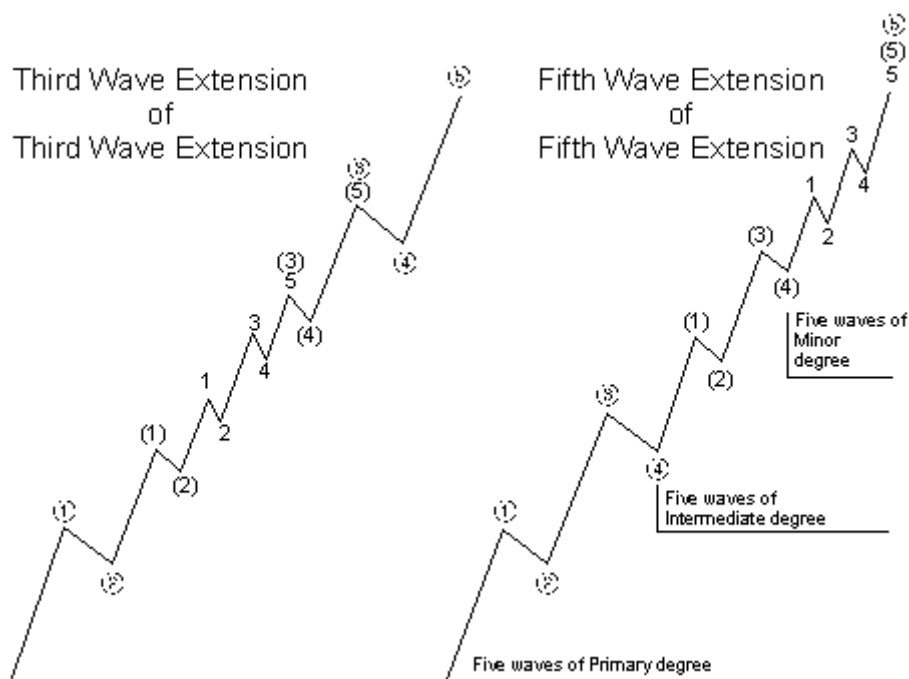


Figure 1-9 Figure 1-10

2.3 Truncation

Elliott used the word "failure" to describe a situation in which the fifth wave does not move beyond the end of the third. We prefer the less connotative term, "truncation," or "truncated fifth." A truncation can usually be verified by noting that the presumed fifth wave contains the necessary five subwaves, as illustrated in Figures 1-11 and 1-12. Truncation often occurs following an extensively strong third wave.

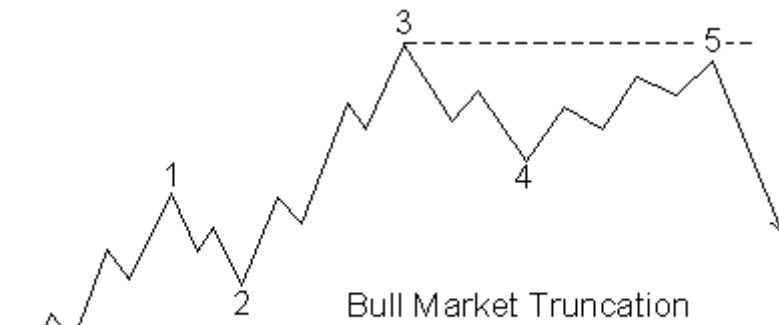


Figure 1-11



Figure 1-12

The U.S. stock market provides two examples of major degree truncated fifths since 1932. The first occurred in October 1962 at the time of the Cuban crisis (see Figure 1-13). It followed the crash that occurred as wave 3. The second occurred at year-end in 1976 (see Figure 1-14). It followed the soaring and broad wave (3) that took place from October 1975 to March 1976.

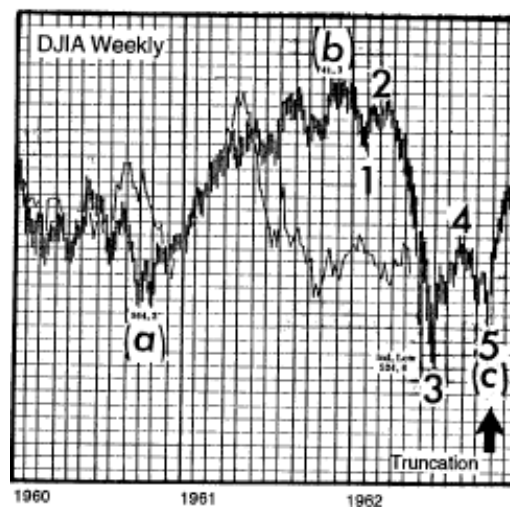


Figure 1-13

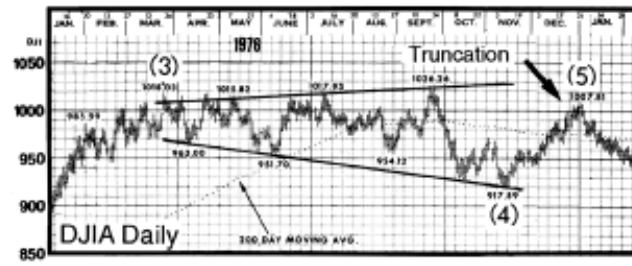


Figure 1-14

Lesson 3

3.1 Diagonal Triangles

A diagonal triangle is a motive pattern yet not an impulse, as it has one or two corrective characteristics. Diagonal triangles substitute for impulses at specific locations in the wave structure. As with impulses, no reactionary subwave fully retraces the preceding actionary subwave, and the third subwave is never the shortest. However, diagonal triangles are the only five-wave structures in the direction of the main trend within which wave four almost always moves into the price territory of (i.e., overlaps) wave one. On rare occasions, a diagonal triangle may end in a truncation, although in our experience such truncations occur only by the slimmest of margins.

Ending Diagonal

An ending diagonal is a special type of wave that occurs primarily in the fifth wave position at times when the preceding move has gone "too far too fast," as Elliott put it. A very small percentage of ending diagonals appear in the C wave position of A-B-C formations. In double or triple threes (to be covered in Lesson 9), they appear only as the *final* "C" wave. In all cases, they are found at the *termination points of larger patterns*, indicating exhaustion of the larger movement.

Ending diagonals take a wedge shape within two converging lines, with each subwave, including waves 1, 3 and 5, subdividing into a "three," which is otherwise a corrective wave phenomenon. The ending diagonal is illustrated in Figures 1-15 and 1-16 and shown in its typical position in larger impulse waves.

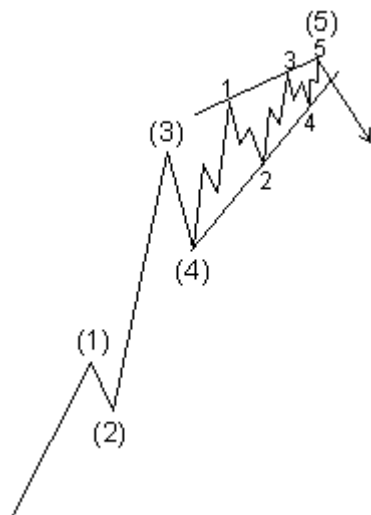


Figure 1-15

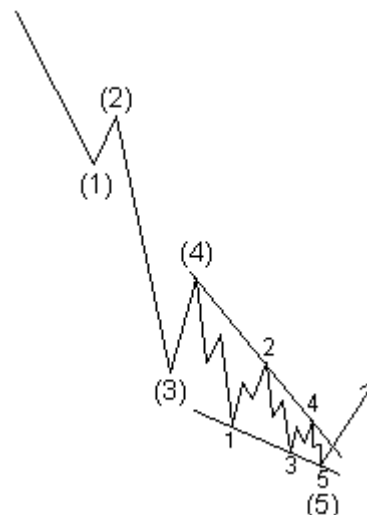


Figure 1-16

We have found one case in which the pattern's boundary lines *diverged*, creating an expanding wedge rather than a contracting one. However, it is unsatisfying analytically in that its third wave was the shortest actionary wave, the entire formation was larger than normal, and another interpretation was *possible*, if not attractive. For these reasons, we do not include it as a valid variation.

3.2 Diagonals

Ending diagonals have occurred recently in Minor degree as in early 1978, in Minute degree as in February-March 1976, and in Subminuette degree as in June 1976. Figures 1-17 and 1-18 show two of these periods, illustrating one upward and one downward "real-life" formation. Figure 1-19 shows our real-life possible expanding diagonal triangle. Notice that in each case, an important change of direction followed.

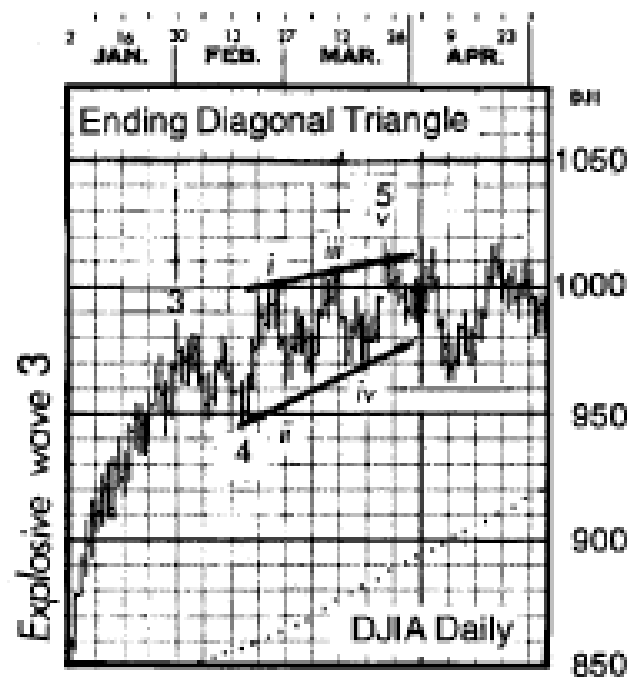


Figure 1-17

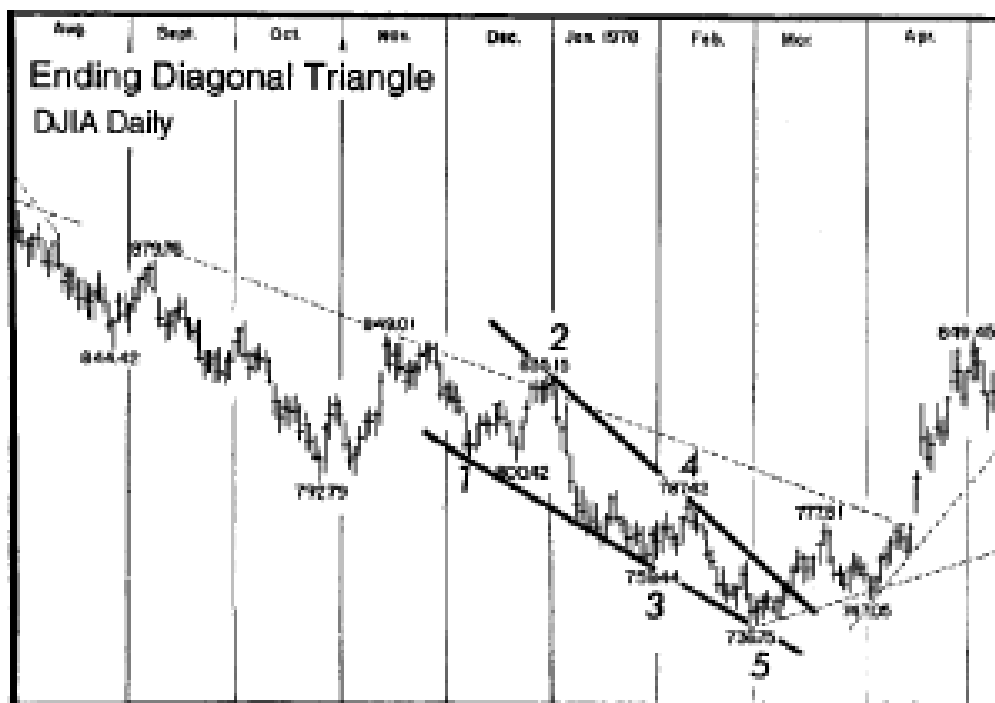


Figure 1-18

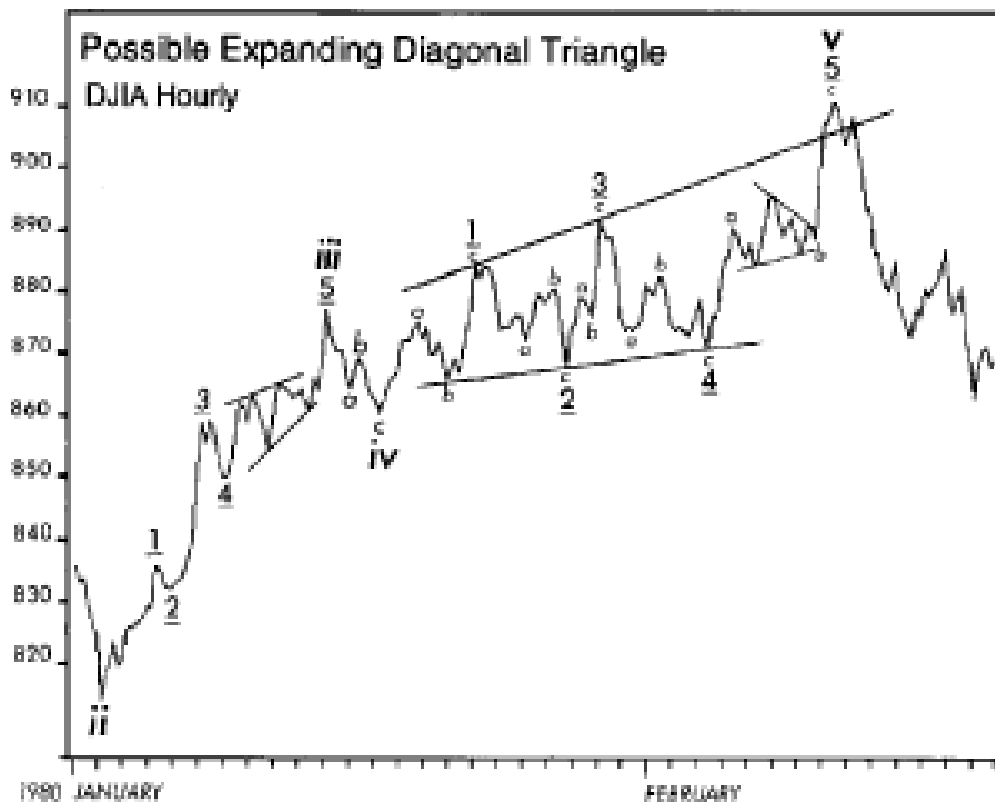


Figure 1-19

Although not so illustrated in Figures 1-15 and 1-16, fifth waves of diagonal triangles often end in a "throw-over," i.e., a brief break of the trendline connecting the end points of waves one and three. Figures 1-17 and 1-19 show real life examples. While volume tends to diminish as a diagonal triangle of small degree progresses, the pattern always ends with a spike of relatively high volume when a throw-over occurs. On rare occasions, the fifth subwave will fall short of its resistance trendline.

A rising diagonal is bearish and is usually followed by a sharp decline retracing at least back to the level where it began. A falling diagonal by the same token is bullish, usually giving rise to an upward thrust.

Fifth wave extensions, truncated fifths and ending diagonal triangles all imply the same thing: *dramatic reversal ahead*. At some turning points, two of these phenomena have occurred together at different degrees, compounding the violence of the next move in the opposite direction.

3.3 Leading Diagonals

When diagonal triangles occur in the wave 5 or C position, they take the 3-3-3-3-3 shape that Elliott described. However, it has recently come to light that a variation on this pattern occasionally appears in the wave 1 position of impulses and in the wave A position of zigzags. The characteristic overlapping of waves 1 and 4 and the convergence of boundary lines into a wedge shape remain as in the ending diagonal triangle. However, the subdivisions are different, tracing out a 5-3-5-3-5 pattern. The structure of this formation (see Figure 1-20) fits the spirit of the Wave Principle in that the five-wave subdivisions in the direction of the larger trend communicate a "continuation" message as opposed to the "termination" implication of the three-wave subdivisions in the ending diagonal. Analysts must be aware of this pattern to avoid mistaking it for a far more common development, a series of first and second waves. The main key to recognizing this pattern is the decided slowing of price change in the fifth subwave relative to the third. By contrast, in developing first and second waves, short term speed typically increases, and breadth (i.e., the number of stocks or subindexes participating) often expands.

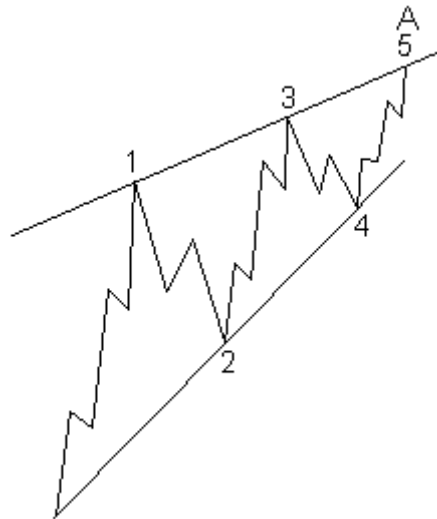


Figure 1-20

Figure 1-21 shows a real life example of a leading diagonal triangle. This pattern was not originally discovered by R.N. Elliott but has appeared enough times and over a long enough period that we are convinced of its validity.

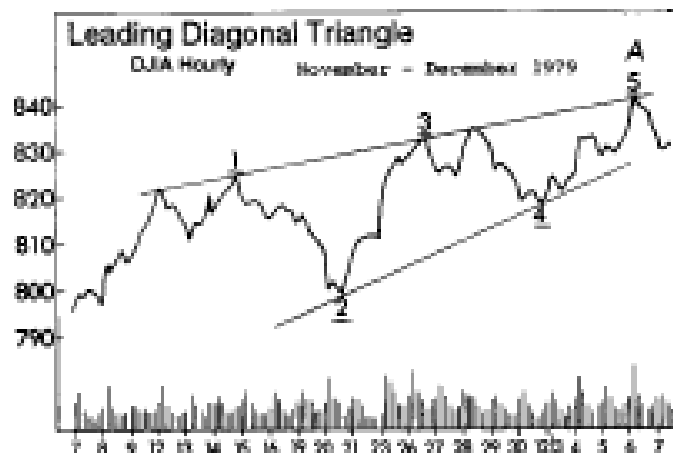


Figure 1-21

3.4 Corrective Waves

Markets move *against* the trend of one greater degree only with a seeming struggle. Resistance from the larger trend appears to prevent a correction from developing a full motive structure. This struggle between the two oppositely trending degrees generally makes corrective waves less clearly identifiable than motive waves, which always flow with comparative ease in the direction of the one larger trend. As another result of this conflict between trends, corrective waves are quite a bit more varied than motive waves. Further, they occasionally increase or decrease in complexity as they unfold so that what are technically subwaves of the same degree can by their complexity or time length appear to be of different degree. For all these reasons, it can be difficult at times to fit corrective waves into recognizable patterns until they are completed and behind us. As the terminations of corrective waves are less predictable than those for motive waves, the Elliott analyst must exercise more caution in his analysis when the market is in a meandering corrective mood than when prices are in a persistently motive trend.

The single most important rule that can be gleaned from a study of the various corrective patterns is that *corrections are never fives*. Only motive waves are fives. For this reason, an initial five-wave movement against the larger trend is never the end of a correction, only part of it. The figures that follow through Lesson 9 of this course should serve to illustrate this point.

Corrective processes come in two styles. *Sharp* corrections angle steeply against the larger trend. *Sideways* corrections, while always producing a net retracement of the preceding wave, typically contain a movement that carries back to or beyond its starting level, thus producing an overall sideways appearance. The discussion of the guideline of alternation in Lesson 10 will explain the reason for noting these two styles.

Specific corrective patterns fall into four main categories:

Zigzags (5-3-5; includes three types: single, double, and triple);

Flats (3-3-5; includes three types: regular, expanded, and running);

Triangles (3-3-3-3-3; four types: three of the contracting variety (ascending, descending, and symmetrical) and one of the expanding variety (reverse symmetrical);

Double threes and *triple threes* (combined structures).

3.5 Zigzags

A *single zigzag* in a bull market is a simple three-wave declining pattern labeled A-B-C. The subwave sequence is 5-3-5, and the top of wave B is noticeably lower than the start of wave A, as illustrated in Figures 1-22 and 1-23.

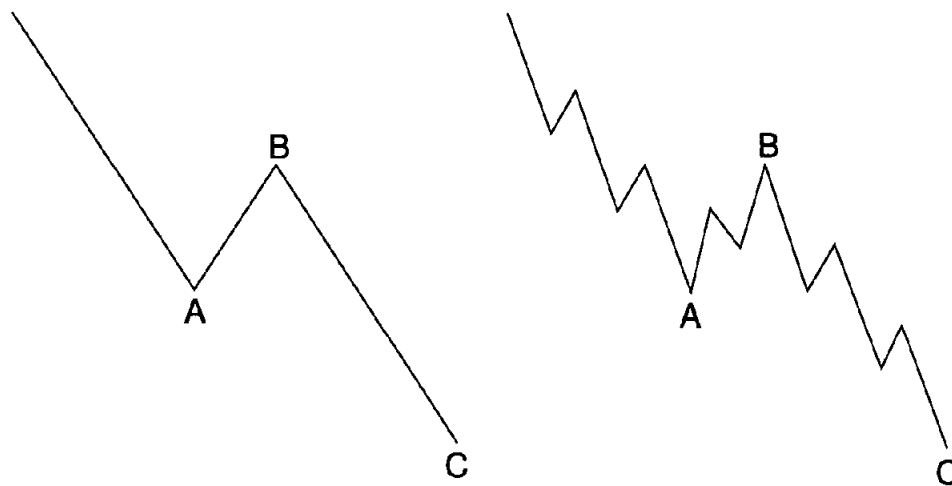


Figure 1-22 Figure 1-23

In a bear market, a zigzag correction takes place in the opposite direction, as shown in Figures 1-24 and 1-25. For this reason, a zigzag in a bear market is often referred to as an inverted zigzag.

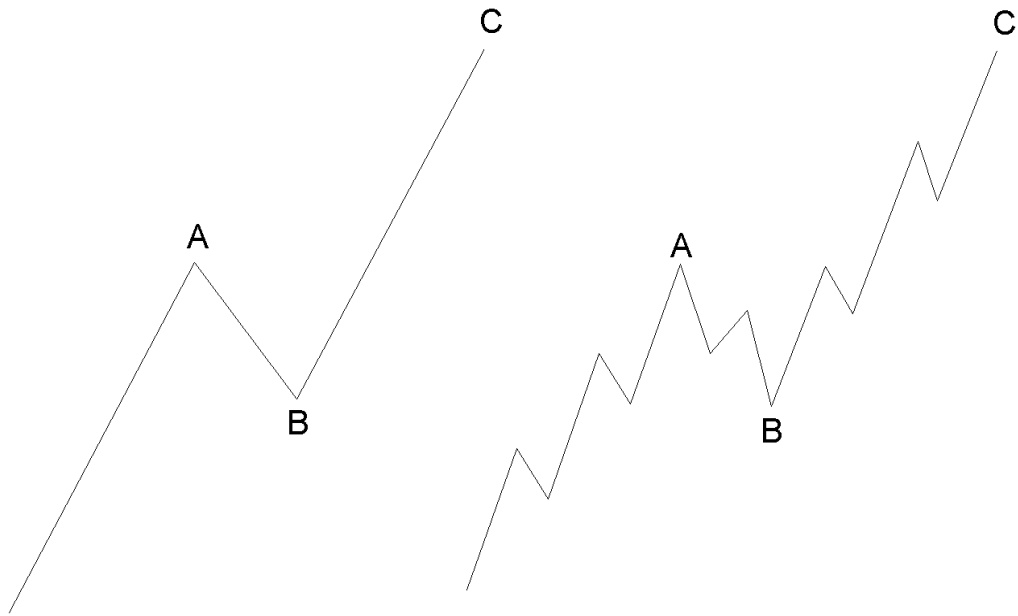


Figure 1-24 Figure 1-25

Occasionally zigzags will occur twice, or at most, three times in succession, particularly when the first zigzag falls short of a normal target. In these cases, each zigzag is separated by an intervening "three," producing what is called a *double zigzag* (see Figure 1-26) or *triple zigzag*. These formations are analogous to the extension of an impulse wave but are less common.

The correction in the Standard and Poor's 500 stock index from

January 1977 to March 1978 (see Figure 1-27) can be labeled as a double zigzag, as can the correction in the Dow from July to October 1975 (see Figure 1-28). Within impulses, second waves frequently sport zigzags, while fourth waves rarely do.

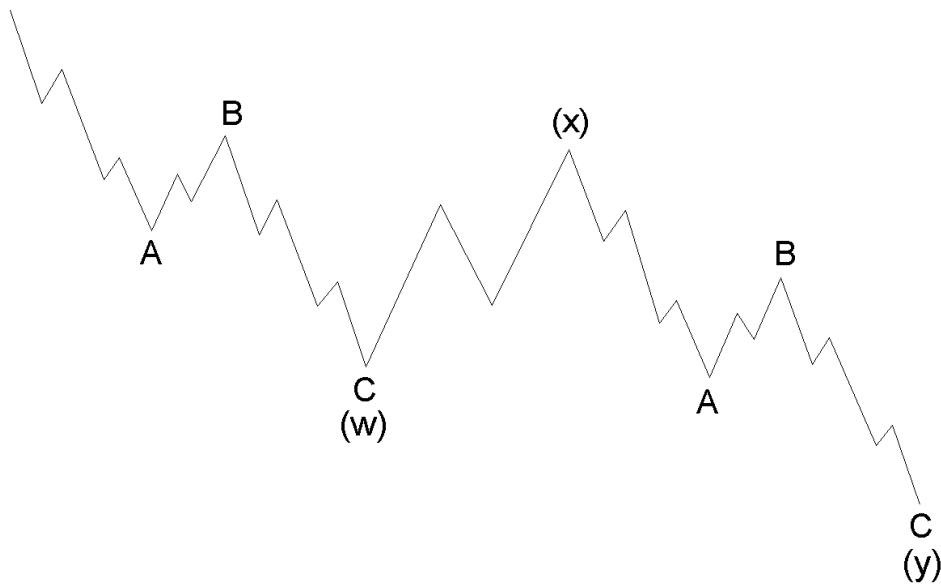


Figure 1-26

Lesson 4

4.1 Flats (3-3-5)

A flat correction differs from a zigzag in that the subwave sequence is 3-3-5, as shown in Figures 1-29 and 1-30. Since the first actionary wave, wave A, lacks sufficient downward force to unfold into a full five waves as it does in a zigzag, the B wave reaction, not surprisingly, seems to inherit this lack of countertrend pressure and terminates near the start of wave A. Wave C, in turn, generally terminates just slightly beyond the end of wave A rather than significantly beyond as in zigzags.

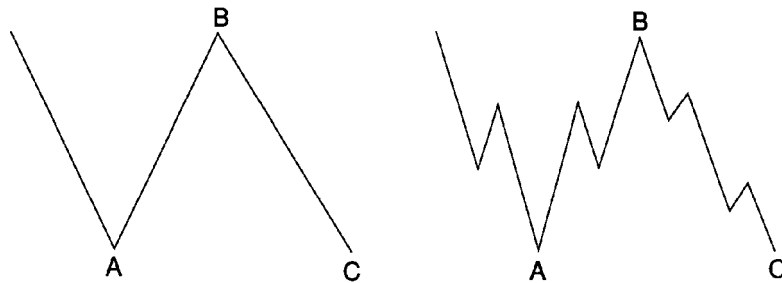


Figure 1-29 Figure 1-30

In a bear market, the pattern is the same but inverted, as shown in Figures 1-31 and 1-32.

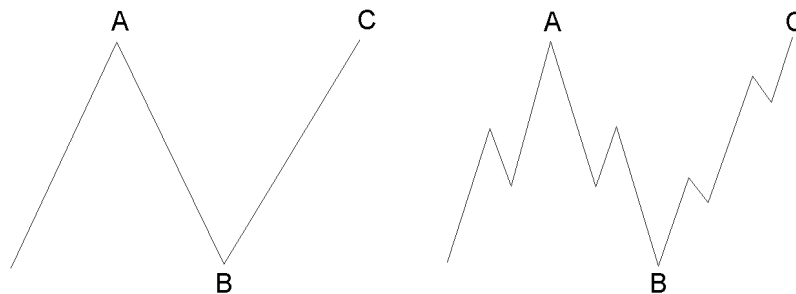


Figure 1-31 Figure 1-32

Flat corrections usually retrace less of preceding impulse waves than do zigzags. They participate in periods involving a strong larger trend and thus virtually always precede or follow extensions. The more powerful the underlying trend, the briefer the flat tends to be. Within impulses, fourth waves frequently sport flats, while second waves do so less commonly.

What might be called "double flats" do occur. However, Elliott categorized such formations as "double threes," a term we discuss in Lesson 9.

The word "flat" is used as a catchall name for any A-B-C correction that subdivides into a 3-3-5. In Elliott literature, however, three types of 3-3-5 corrections have been identified by differences in their overall shape. In a *regular* flat correction, wave B terminates about at the level of the beginning of wave A, and wave C terminates a slight bit past the end of wave A, as we have shown in Figures 1-29 through 1-32. Far more common, however, is the variety called an *expanded* flat, which contains a price extreme beyond that of the preceding impulse wave. Elliott called this variation an "irregular" flat, although the word is inappropriate as they are actually far more common than "regular" flats.

4.2 Expanded Flats

In expanded flats, wave B of the 3-3-5 pattern terminates beyond the starting level of wave A, and wave C ends more substantially beyond the ending level of wave A, as shown for bull markets in Figures 1-33 and 1-34 and bear markets in Figures 1-35 and 1-36. The formation in the DJIA from August to November 1973 was an expanded flat correction of this type in a bear market, or an "inverted expanded flat" (see Figure 1-37).

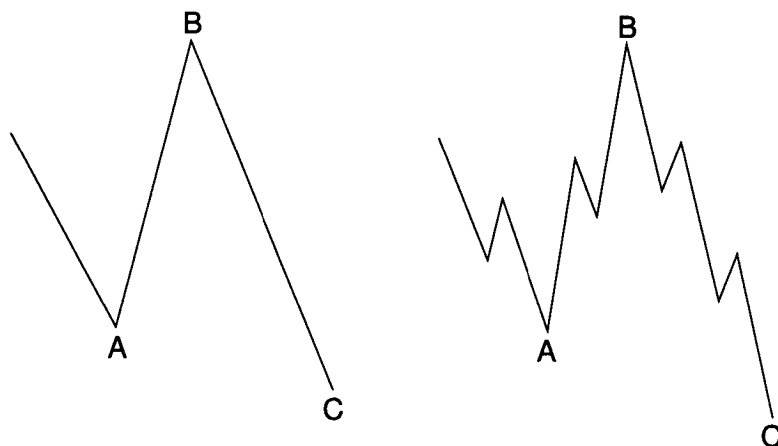


Figure 1-33 Figure 1-34

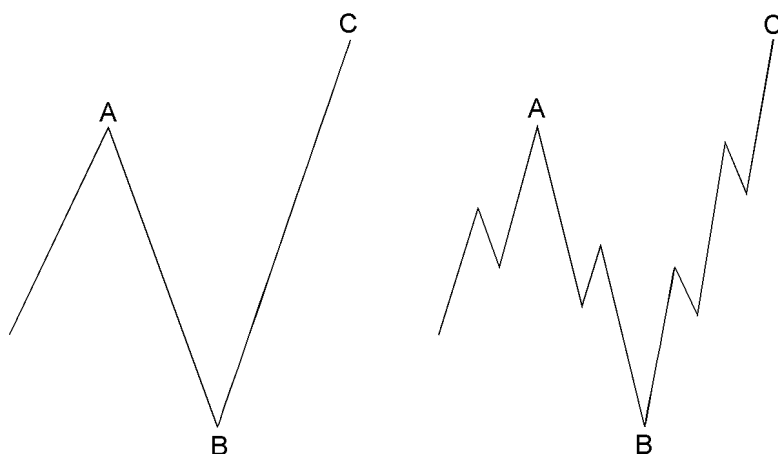


Figure 1-35 Figure 1-36

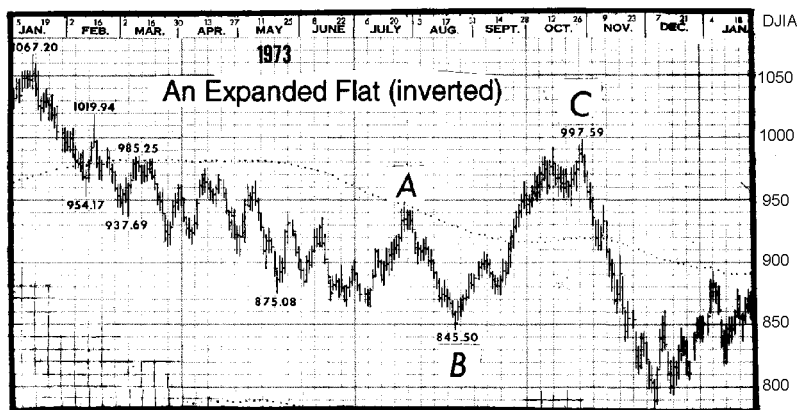


Figure 1-37

In a rare variation on the 3-3-5 pattern, which we call a *running flat*, wave B terminates well beyond the beginning of wave A as in an expanded flat, but wave C fails to travel its full distance, falling short of the level at which wave A ended, as in Figures 1-38 through 1-41. Apparently in this case, the forces in the direction of the larger trend are so powerful that the pattern becomes skewed in that direction. It is always important, but particularly when concluding that a running flat has taken place, that the internal subdivisions adhere to Elliott's rules. If the supposed B wave, for instance, breaks down into five waves rather than three, it is more likely the first wave up of the impulse of next higher degree. The power of adjacent impulse waves is important in recognizing running corrections, which tend to occur only in strong and fast markets. We must issue a warning, however. There are hardly any examples of this type of correction in the price record. Never label a correction prematurely this way, or you'll find yourself wrong nine times out of ten. Running *triangles*, in contrast, are much more common, as we'll see in Lesson 8.

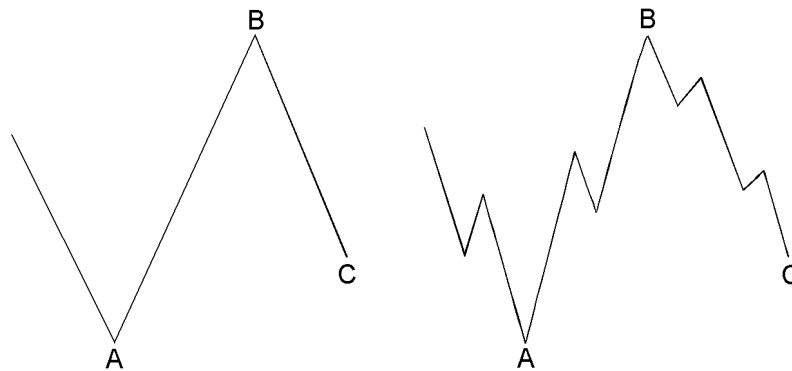
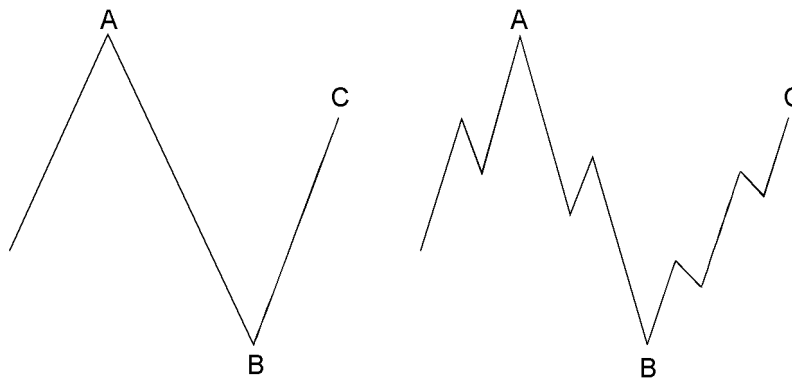


Figure 1-38 Figure 1-39



4.3 Triangles

Triangles appear to reflect a balance of forces, causing a sideways movement that is usually associated with decreasing volume and volatility. Triangles contain five overlapping waves that subdivide 3-3-3-3-3 and are labeled a-b-c-d-e. A triangle is delineated by connecting the termination points of waves a and c, and b and d. Wave e can undershoot or overshoot the a-c line, and in fact, our experience tells us that it happens more often than not.

There are two varieties of triangles: contracting and expanding. Within the contracting variety, there are three types: symmetrical, ascending, and descending, as illustrated in Figure 1-42. There are no variations on the rarer expanding triangle. It always appears as depicted in Figure 1-42, which is why Elliott termed it a "reverse symmetrical" triangle.

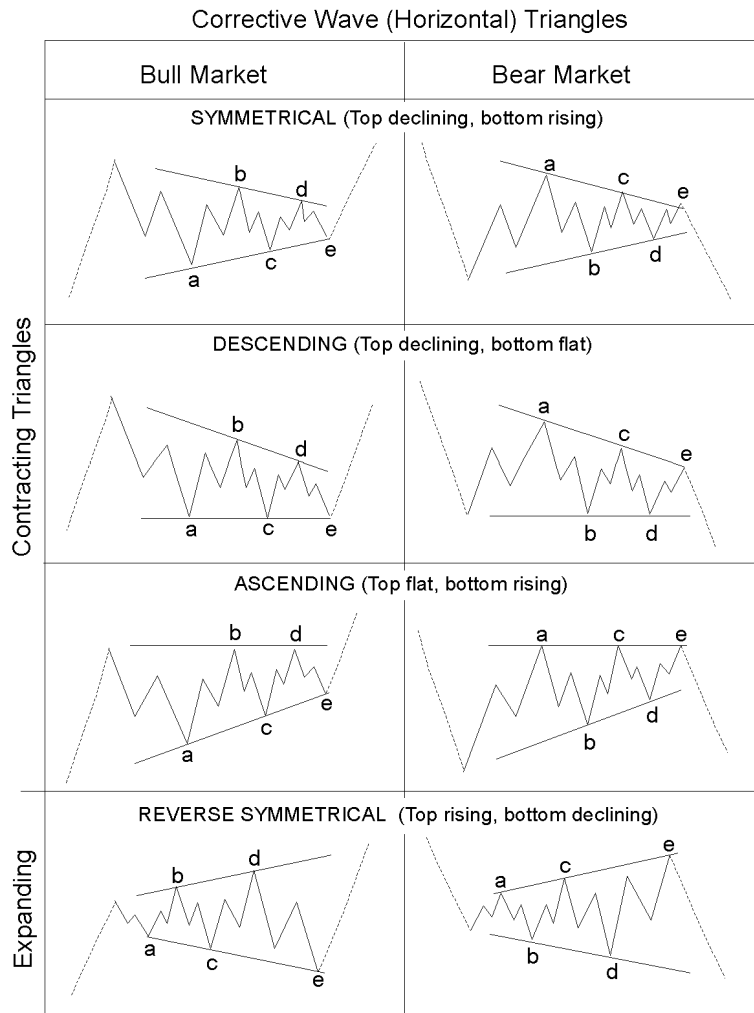


Figure 1-42

Figure 1-42 depicts contracting triangles as taking place within the area of preceding price action, in what may be termed *regular* triangles. However, it is extremely common for wave b of a contracting triangle to exceed the start of wave a in what may be termed a *running* triangle, as shown in Figure 1-43. Despite their sideways appearance, *all* triangles, including running triangles, effect a net retracement of the preceding wave at wave e's end.

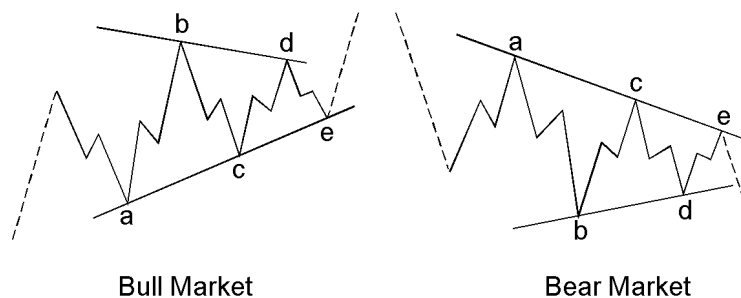


Figure 1-43

4.4 Triangle Examples

There are several real life examples of triangles in the charts in this course. As you will notice, most of the subwaves in a triangle are zigzags, but sometimes one of the subwaves (usually wave c) is more complex than the others and can take the shape of a regular or expanded flat or multiple zigzag. In rare cases, one of the sub-waves (usually wave e) is itself a triangle, so that the entire pattern protracts into nine waves. Thus, triangles, like zigzags, occasionally display a development that is analogous to an extension. One example occurred in silver from 1973 through 1977 (see Figure 1-44).

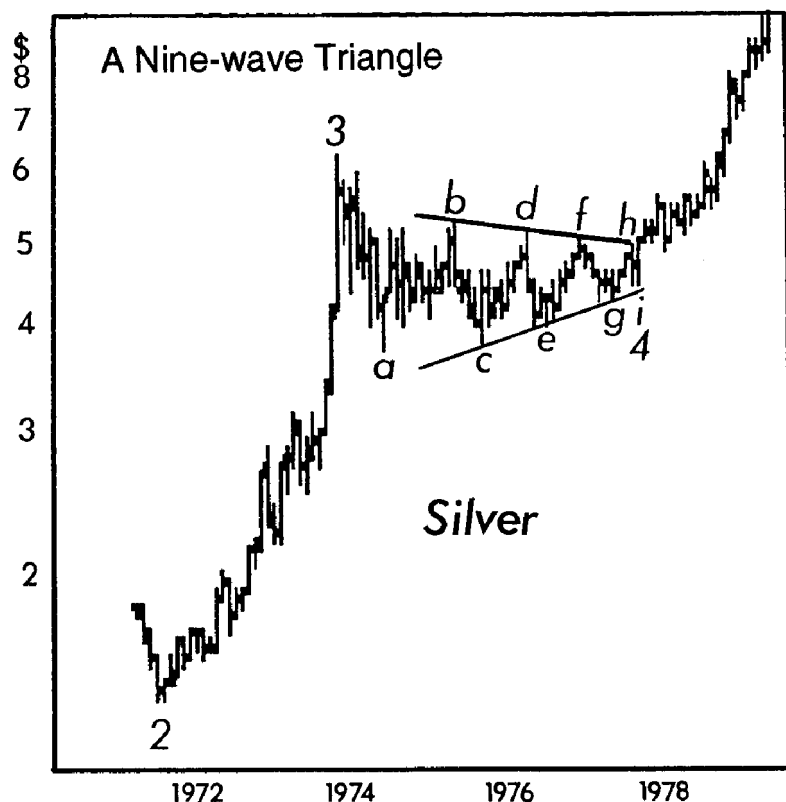


Figure 1-44

Although upon extremely rare occasions a second wave in an impulse appears to take the form of a triangle, triangles nearly always occur in positions *prior* to the final actionary wave in the pattern of one larger degree, i.e., as wave four in an impulse, wave B in an A-B-C, or the final wave X in a double or triple zig-zag or combination (to be shown in Lesson 9). A triangle may also occur as the final actionary pattern in a corrective combination, as discussed in Lesson 9, although even then it always precedes the final actionary wave in the pattern of one larger degree than the corrective combination.

In the stock market, when a triangle occurs in the fourth wave position, wave five is sometimes swift and travels approximately the distance of the widest part of the triangle. Elliott used the word "thrust" in referring to this swift, short motive wave following a triangle. The thrust is usually an impulse but can be an ending diagonal. In powerful markets, there is no thrust, but instead a prolonged fifth wave. So if a fifth wave following a triangle pushes past a normal thrust measurement, it is signaling a likely protracted wave. Post-triangle advancing impulses in commodities at degrees above Intermediate are usually the longest wave in the sequence, as explained in Lesson 29.

On the basis of our experience with triangles, as the example in Figure 3-15 illustrates, we propose that often the time at which the boundary lines of a contracting triangle reach an apex coincides exactly with a turning point in the market. Perhaps the frequency of this occurrence would justify its inclusion among the guidelines associated with the Wave Principle.

The term "horizontal" as applied to triangles refers to these corrective triangles in general, as opposed to the term "diagonal," which refers to those motive triangular formations discussed in Lesson 5. Thus, the terms "horizontal triangle" and "diagonal triangle" denote these specific forms under the Wave Principle. The simpler terms "triangle" and "wedge" may be substituted, but keep in mind that technical chart readers have long used these terms to communicate less specifically subdivided forms defined only by overall shape. Having separate terms can be useful.

Lesson 5

5.1 Corrective Combinations

Double and Triple Threes

Elliott called sideways combinations of corrective patterns "double threes" and "triple threes." While a single three is any zigzag or flat, a triangle is an allowable final component of such combinations and in this context is called a "three." A double or triple three, then, is a combination of simpler types of corrections, including the various types of zigzags, flats and triangles. Their occurrence appears to be the flat correction's way of extending sideways action. As with double and triple zigzags, each simple corrective pattern is labeled W, Y and Z. The reactionary waves, labeled X, can take the shape of any corrective pattern but are most commonly zigzags.

Combinations of threes were labeled differently by Elliott at different times, although the illustrative pattern always took the shape of two or three juxtaposed flats, as shown in Figures 1-45 and 1-46. However, the component patterns more commonly alternate in form. For example, a flat followed by a triangle is a more typical type of double three, as illustrated in Figure 1-47.

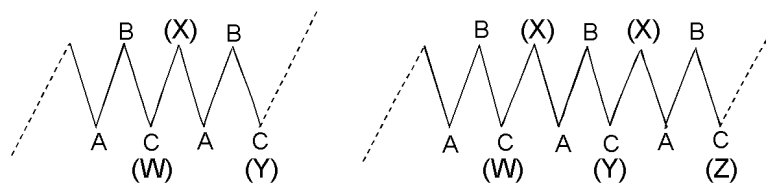


Figure 1-45 Figure 1-46

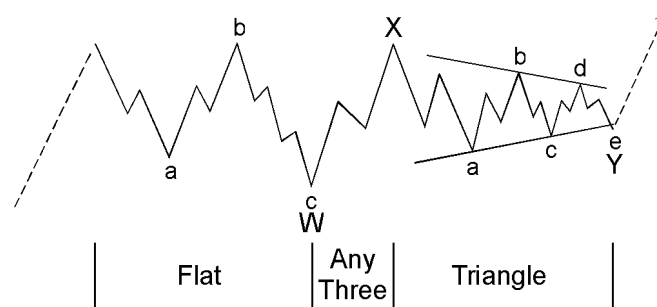


Figure 1-47

A flat followed by a zigzag is another example, as shown in Figure 1-48. Naturally, since the figures in this section depict corrections in bull markets, they need only be inverted to observe them as upward corrections in bear markets.

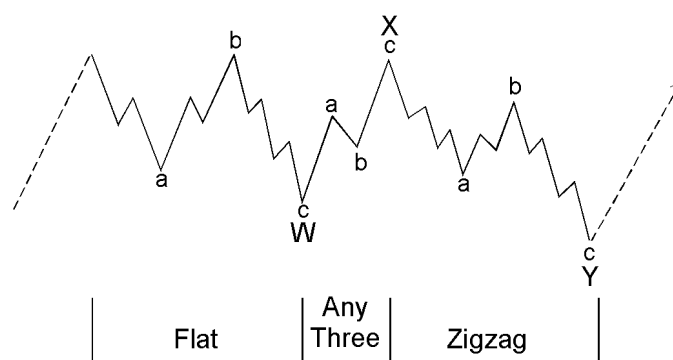


Figure 1-48

For the most part, double threes and triple threes are horizontal in character. Elliott indicated that the entire formations could slant against the larger trend, although we have never found this to be the case. One reason is that there never appears to be more than one zigzag in a combination. Neither is there more than one triangle. Recall that triangles occurring alone precede the final movement of a larger trend. Combinations appear to recognize this character and sport triangles only as the final wave in a double or triple three.

5.2 Guidelines of Wave Formation

Although different in that their angle of trend is sharper than the sideways trend of combinations, double and triple zigzags can be characterized as non-horizontal combinations, as Elliott seemed to suggest in *Nature's Law*. However, double and triple threes are different from double and triple zigzags, not only in their angle but in their goal. In a double or triple zigzag, the first zigzag is rarely large enough to constitute an adequate price correction of the preceding wave. The doubling or tripling of the initial form is typically necessary to create an adequately sized *price* retracement. In a combination, however, the first simple pattern often constitutes an adequate price correction. The doubling or tripling appears to occur mainly to extend the *duration* of the corrective process after price targets have been substantially met. Sometimes additional time is needed to reach a channel line or achieve a stronger kinship with the other correction in an impulse wave. As the consolidation continues, the attendant psychology and fundamentals extend their trends accordingly.

As this section makes clear, there is a qualitative difference between the number series **3** + 4 + 4 + 4, etc., and the series **5** + 4 + 4 + 4, etc. Notice that while impulse waves have a total count of 5, with extensions leading to 9, 13 or 17 waves, and so on, corrective waves have a count of 3, with combinations leading to 7 or 11 waves, and so on. Triangles appear to be an exception, although they can be counted as one would a triple three, totaling 11 waves. Thus, if an internal count is unclear, the analyst can sometimes reach a reasonable conclusion merely by counting waves. A count of 9, 13 or 17 with few overlaps, for instance, is likely motive, while a count of 7, 11 or 15 with numerous overlaps is likely corrective. The main exceptions are diagonal triangles of both types, which are hybrids of motive and corrective forces.

Orthodox Tops and Bottoms

Sometimes a pattern's end differs from the associated price extreme. In such cases, the end of the pattern is called the "orthodox" top or bottom in order to differentiate it from the actual price high or low that occurs intra-pattern. For example, in Figure 1-11, the end of wave 5 is the orthodox top despite the fact that wave 3 registered a higher price. In Figure 1-12, the end of wave 5 is the orthodox bottom. In Figures 1-33 and 1-34, the starting point of wave A is the orthodox top of the preceding bull market despite the higher high of wave B. In Figure 1-47, the end of wave Y is the orthodox bottom of the bear market even though the price low occurs at the end of wave W.

This concept is important primarily because a successful analysis always depends upon a proper labeling of the patterns. Assuming falsely that a particular price extreme is the correct starting point for wave labeling can throw analysis off for some time, while being aware of the requirements of wave form will keep you on track. Further, when applying the forecasting concepts that will be introduced in Lessons 20 through 25, the length and duration of a wave are typically determined by measuring from and projecting orthodox ending points.

5.3 Reconciling Function and mode

In Lessons 3 and 4, we described the two functions waves may perform (action and reaction), as well as the two modes of structural development (motive and corrective) that they undergo. Now that we have reviewed all types of waves, we can summarize their labels as follows:

— The labels for actionary waves are 1, 3, 5, A, C, E, W, Y and Z.

— The labels for reactionary waves are 2, 4, B, D and X.

As stated earlier, *all* reactionary waves develop in corrective mode, and *most* actionary waves develop in motive mode. The preceding sections have described which actionary waves develop in corrective mode. They are:

- waves 1, 3 and 5 in an ending diagonal,
- wave A in a flat correction,
- waves A, C and E in a triangle,
- waves W and Y in double zigzags and double corrections,
- wave Z in triple zigzags and triple corrections.

Because the waves listed above are actionary in relative direction yet develop in corrective mode, we term them "actionary corrective" waves.

As far as we know, we have listed all wave formations that can occur in the price movement of the broad stock market averages. Under the Wave Principle, no other formations than those listed here will occur. Indeed, since the hourly readings are a nearly perfectly matched filter for detailing waves of Subminuette degree, the authors can find no examples of waves above the Subminuette degree that cannot be counted satisfactorily by the Elliott method. In fact, Elliott Waves of much smaller degree than Subminuette are revealed by computer generated charts of minute-by-minute transactions. Even the few data points (transactions) per unit of time at this low a degree are enough to reflect accurately the Wave Principle of human behavior by recording the rapid shifts in psychology occurring in the "pits" and on the exchange floor. All rules (which were covered in Lessons 1 through 9) and guidelines (which are covered in Lessons 1 through 15) fundamentally apply to actual market mood, not its recording *per se* or lack thereof. Its clear manifestation requires free market pricing. When prices are fixed by government edict, such as those for gold and silver for half of the twentieth century, waves restricted by the edict are not allowed to register. When the available price record differs from what might have existed in a free market, rules and guidelines must be considered in that light. In the long run, of course, markets always win out over edicts, and edict enforcement is only possible if the mood of the market allows it. All rules and guidelines presented in this course presume that your price record is accurate. Now that we have presented the rules and rudiments of wave formation, we can move on to some of the guidelines for successful analysis under the Wave Principle.

5.4 The Guidelines of Alternation

The guidelines presented in Lessons 10-15 are discussed and illustrated in the context of a bull market. Except where specifically excluded, they apply equally in bear markets, in which context the illustrations and implications would be inverted.

Alternation

The guideline of alternation is very broad in its application and warns the analyst always to expect a difference in the next expression of a similar wave. Hamilton Bolton said,

The writer is *not* convinced that alternation is *inevitable* in types of waves in larger formations, but there are frequent enough cases to suggest that one should look for it rather than the contrary.

Although alternation does not say precisely what is going to happen, it gives valuable notice of what *not* to expect and is therefore useful to keep in mind when analyzing wave formations and assessing future possibilities. It primarily instructs the analyst not to assume, as most people tend to do, that because the last market cycle behaved in a certain manner, this one is sure to be the same. As "contrarians" never cease to point out, the day that most investors "catch on" to an apparent habit of the market is the day it will change to one completely different. However, Elliott went further in stating that, in fact, alternation was virtually a law of markets.

Alternation Within Impulses

If wave two of an impulse is a sharp correction, expect wave four to be a sideways correction, and vice versa. Figure 2-1 shows the most characteristic breakdowns of impulse waves, both up and down, as suggested by the guideline of alternation. Sharp corrections never include a new price extreme, i.e., one that lies beyond the orthodox end of the preceding impulse wave. They are almost always zigzags (single, double or triple); occasionally they are double threes that *begin* with a zigzag. Sideways corrections include flats, triangles, and double and triple corrections. They usually include a new price extreme, i.e., one that lies beyond the orthodox end of the preceding impulse wave. In rare cases, a regular triangle (one that does not include a new price extreme) in the fourth wave position will take the place of a sharp correction and alternate with another type of sideways pattern in the second wave position. The idea of alternation within impulses can be summarized by saying that one of the two corrective processes will contain a move back to or beyond the end of the preceding impulse, and the other will not.

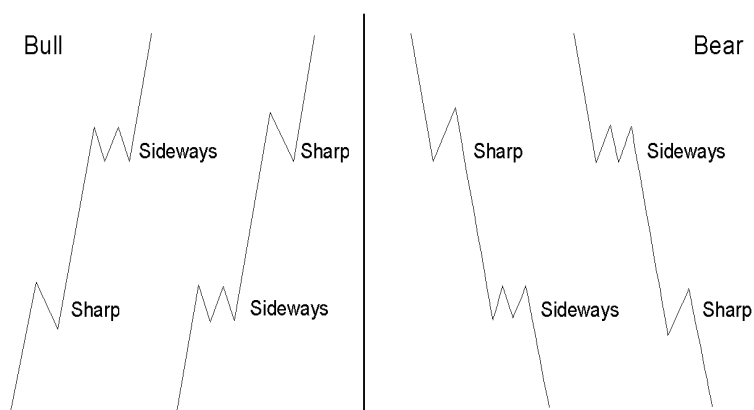
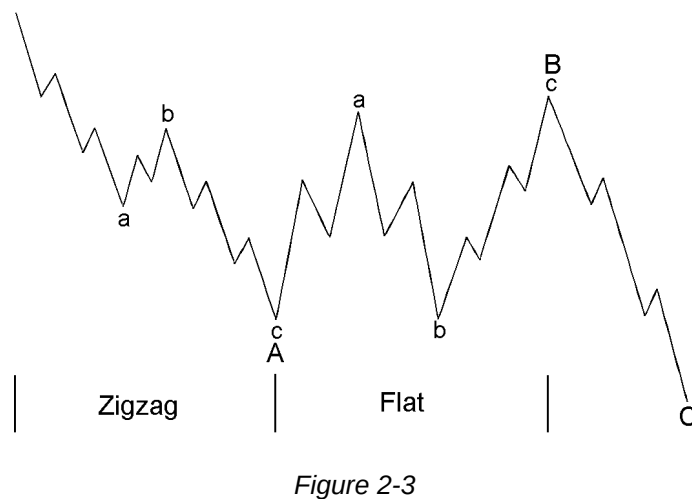
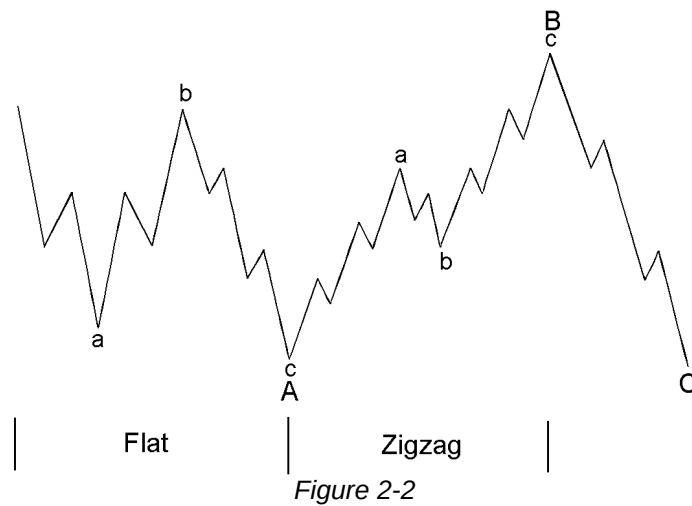


Figure 2-1

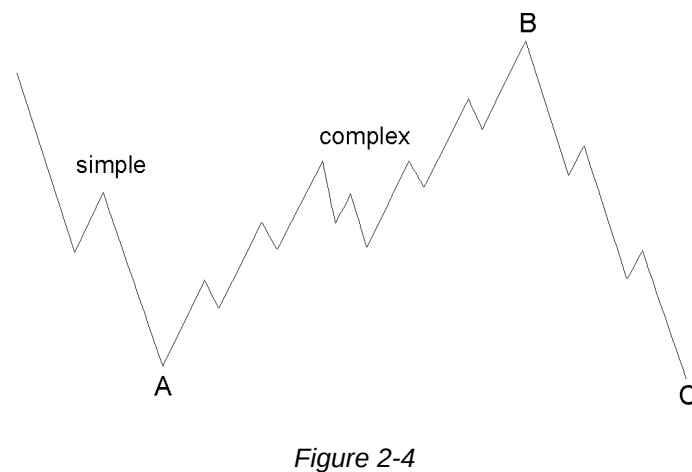
Diagonal triangles do not display alternation between subwaves 2 and 4. Typically they are both zigzags. Extensions are an expression of alternation, as the motive waves alternate their lengths. Typically the first is short, the third is extended, and the fifth is short again. Extensions, which normally occur in wave 3, sometimes occur in wave 1 or 5, another manifestation of alternation.

5.5 Alternation within Corrective Waves

If a large correction begins with a flat a-b-c construction for wave A, expect a zigzag a-b-c formation for wave B (see Figure 2-2), and vice versa (see Figure 2-3). With a moment's thought, it is obvious that this occurrence is sensible, since the first illustration reflects an upward bias in both subwaves while the second reflects a downward bias.



Quite often, if a large correction begins with a simple a-b-c zigzag for wave A, wave B will stretch out into a more intricately subdivided a-b-c zigzag to achieve a type of alternation, as in Figure 2-4. Sometimes wave C will be yet more complex, as in Figure 2-5. The reverse order of complexity is somewhat less common.



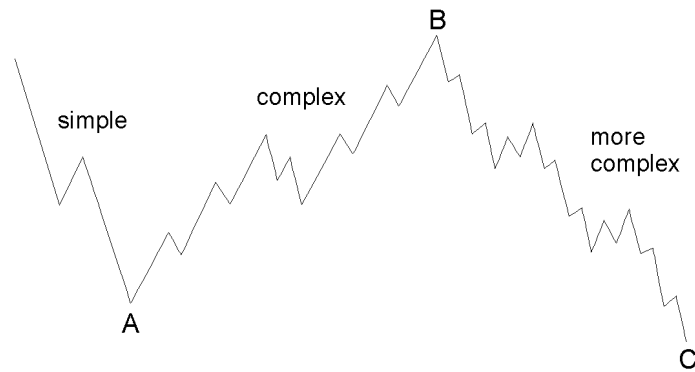


Figure 2-5

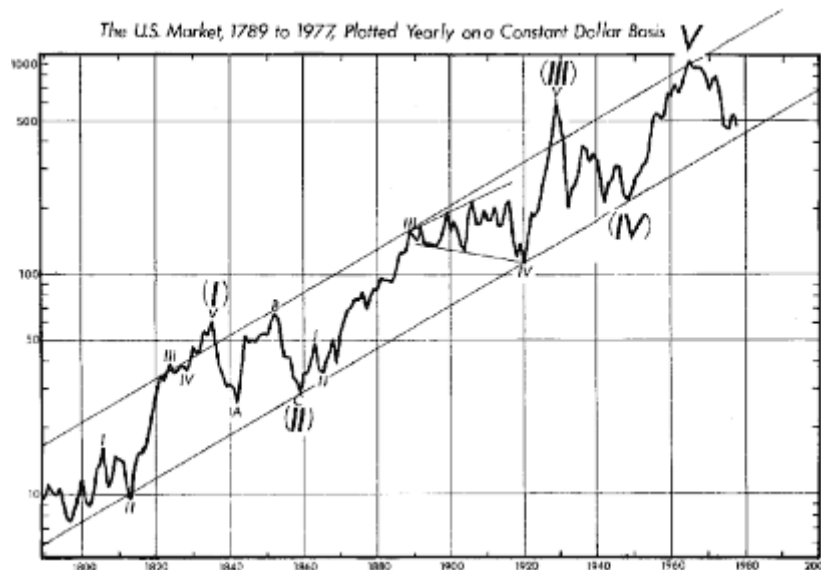
5.6 Forecasting Corrective Waves

Depth of Corrective Waves (Bear Market Limitations)

No market approach other than the Wave Principle gives as satisfactory an answer to the question, "How far down can a bear market be expected to go?" The primary guideline is that corrections, especially when they themselves are fourth waves, tend to register their maximum retracement within the span of travel of the previous fourth wave of one lesser degree, most commonly near the level of its terminus.

Example #1: The 1929-1932 Bear Market

The chart of stock prices adjusted to constant dollars developed by the Foundation for the Study of Cycles shows a contracting triangle as wave (IV). Its lows bottom within the area of the previous fourth wave of Cycle degree, an expanding triangle (see chart below).



Example #2: The 1942 Bear Market Low

In this case, the Cycle degree wave II bear market from 1937 to 1942, a zigzag, terminates within the area of Primary wave [4] of the bull market from 1932 to 1937 (see Figure 5-3).

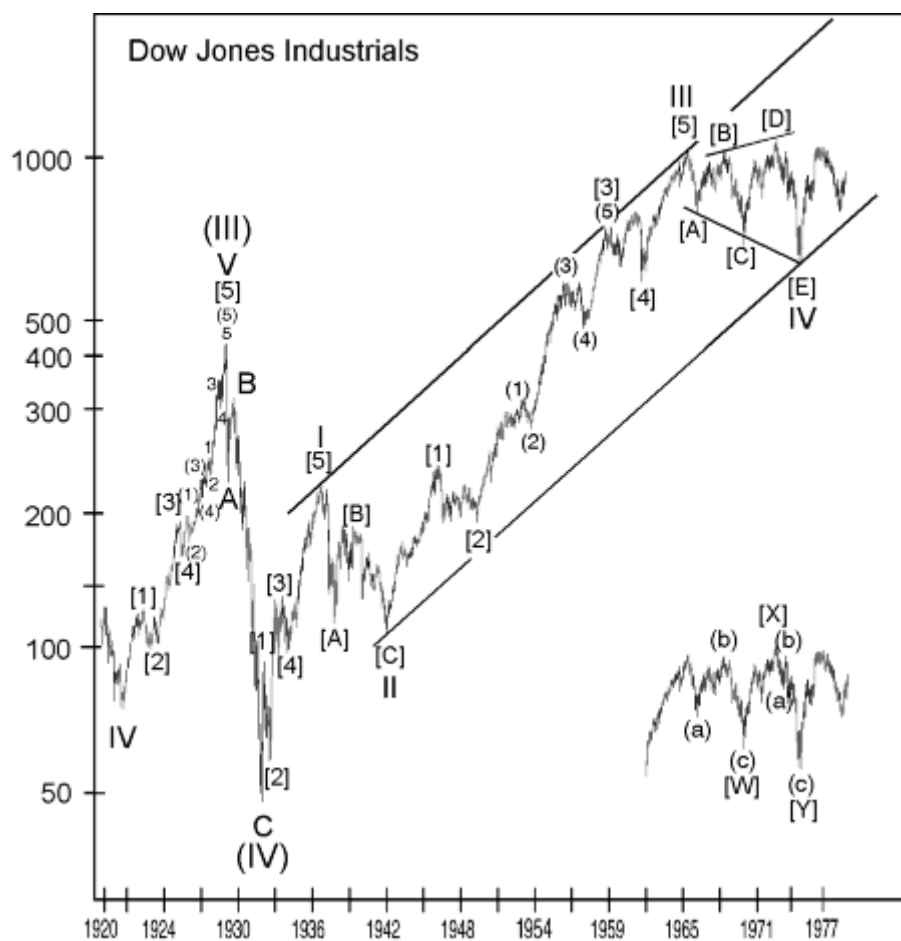


Figure 5-3

Example #3: The 1962 Bear Market Low

The wave [4] plunge in 1962 brought the averages down to just above the 1956 high of the five-wave Primary sequence from 1949 to 1959. Ordinarily, the bear would have reached into the zone of wave (4), the fourth wave correction within wave [3]. This narrow miss nevertheless illustrates why this guideline is not a rule. The preceding strong third wave extension and the shallow A wave and strong B wave within [4] indicated strength in the wave structure, which carried over into the moderate net depth of the correction (see Figure 5-3).

5.7 Wave Extensions

Example #4: The 1974 Bear Market Low

The final decline into 1974, ending the 1966-1974 Cycle degree wave IV correction of the entire wave III rise from 1942, brought the averages down to the area of the previous fourth wave of lesser degree (Primary wave [4]). Again, Figure 5-3 shows what happened.

Our analysis of small degree wave sequences over the last twenty years further validates the proposition that the usual limitation of any bear market is the travel area of the preceding fourth wave of one lesser degree, particularly when the bear market in question is itself a fourth wave. However, in a clearly reasonable modification of the guideline, it is often the case that if the *first* wave in a sequence extends, the correction following the fifth wave will have as a typical limit the bottom of the *second* wave of lesser degree. For example, the decline into March 1978 in the DJIA bottomed

exactly at the low of the second wave in March 1975, which followed an extended first wave off the December 1974 low.

On occasion, flat corrections or triangles, particularly those following extensions (see Example #3), will barely fail to reach into the fourth wave area. Zigzags, on occasion, will cut deeply and move down into the area of the second wave of lesser degree, although this almost exclusively occurs when the zigzags are themselves second waves. "Double bottoms" are sometimes formed in this manner.

Behavior Following Fifth Wave Extensions

The most important empirically derived rule that can be distilled from our observations of market behavior is that when the fifth wave of an advance is an extension, the ensuing correction will be sharp and find support at the level of the low of wave two of the extension. Sometimes the correction will end there, as illustrated in Figure 2-6. Although a limited number of real life examples exist, the precision with which "A" waves have reversed at the level of the low of wave two of the preceding fifth wave extension is remarkable. Figure 2-7 is an illustration involving an expanded flat correction. (For future reference, please make a note of two real-life examples that we will show in charts of upcoming lessons. An example involving a zigzag can be found in Figure 5-3 at the low of wave [a] of II, and an example involving an expanded flat can be found in Figure 2-16 at the low of wave a of A of 4. As you will see in Figure 5-3, wave A of (IV) bottoms near wave (2) of [5], which is an extension within wave V from 1921 to 1929.)

Since the low of the second wave of an extension is commonly in or near the price territory of the immediately preceding fourth wave of one larger degree, this guideline implies behavior similar to that for the preceding guideline. It is notable for its *precision*, however. Additional value is provided by the fact that fifth wave extensions are typically followed by *swift* retracements. Their occurrence, then, is an advance warning of a dramatic reversal to a specific level, a powerful combination of knowledge. This guideline does not apply separately to fifth wave extensions *of* fifth wave extensions.

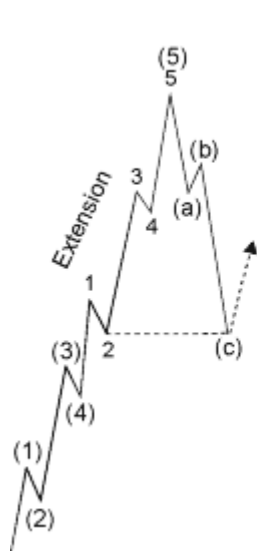


Figure 2-6

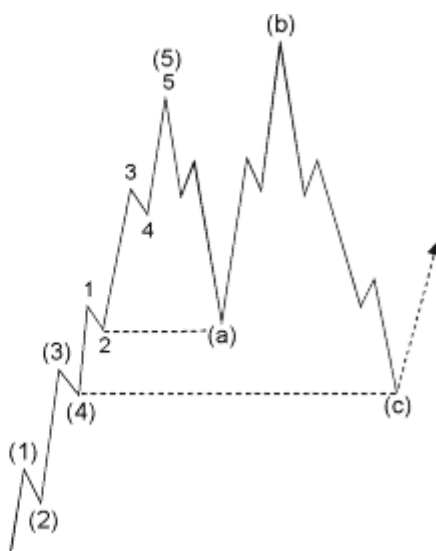


Figure 2-7

Lesson 6

6.1 Channeling

Wave Equality

One of the guidelines of the Wave Principle is that two of the motive waves in a five-wave sequence will tend toward equality in time and magnitude. This is generally true of the two non-extended waves when one wave is an extension, and it is especially true if the third wave is the extension. If perfect equality is lacking, a .618 multiple is the next likely relationship (the use of ratios is covered in Lessons 16-25).

When waves are larger than Intermediate degree, the price relationships usually must be stated in percentage terms. Thus, within the entire extended Cycle wave advance from 1942 to 1966, we find that Primary wave [1] traveled 120 points, a gain of 129%, in 49 months, while Primary wave [5] traveled 438 points, a gain of 80% (.618 times the 129% gain), in 40 months (see Figure 5-3), far different from the 324% gain of the third Primary wave, which lasted 126 months.

When the waves are of Intermediate degree or less, the price equality can usually be stated in arithmetic terms, since the percentage lengths will also be nearly equivalent. Thus, in the year-end rally of 1976, we find that wave 1 traveled 35.24 points in 47 market hours while wave 5 traveled 34.40 points in 47 market hours. The guideline of equality is often extremely accurate.

Charting the Waves

A. Hamilton Bolton always kept an "hourly close" chart, i.e., one showing the end-of-hour prices, as do the authors. Elliott himself certainly followed the same practice, since in *The Wave Principle* he presents an hourly chart of stock prices from February 23 to March 31, 1938. Every Elliott Wave practitioner, or anyone interested in the Wave Principle, will find it instructive and useful to plot the hourly fluctuations of the DJIA, which are published by *The Wall Street Journal* and *Barron's*. It is a simple task that requires only a few minutes' work a week. Bar charts are fine but can be misleading by revealing fluctuations that occur near the time changes for each bar but not those that occur within the time for the bar. Actual print figures must be used on all plots. The so-called "opening" and "theoretical intraday" figures published for the Dow averages are statistical inventions that do not reflect the averages at any particular moment. Respectively, these figures represent a sum of the opening prices, which can occur at different times, and of the daily highs or lows of each individual stock in the average regardless of the time of day each extreme occurs.

The foremost aim of wave classification is to determine where prices are in the stock market's progression. This exercise is easy as long as the wave counts are clear, as in fast-moving, emotional markets, particularly in impulse waves, when minor movements generally unfold in an uncomplicated manner. In these cases, short term charting is necessary to view all subdivisions. However, in lethargic or choppy markets, particularly in corrections, wave structures are more likely to be complex and slow to develop. In these cases, longer term charts often effectively condense the action into a form that clarifies the pattern in progress. With a proper reading of the Wave Principle, there are times when sideways trends can be forecasted (for instance, for a fourth wave when wave two is a zigzag). Even when anticipated, though, complexity and lethargy are two of the most frustrating occurrences for the analyst. Nevertheless, they are part of the reality of the market and must be taken into account. The authors highly recommend that during such periods you take some time off from the market to enjoy the fruits of your hard work. You can't "wish" the market into action; it isn't listening. When the market rests, do the same.

The correct method for tracking the stock market is to use semilogarithmic chart paper, since the market's history is sensibly related only on a percentage basis. The investor is concerned with percentage gain or loss, not the number of points traveled in a market average. For instance, ten points in the DJIA in 1980 meant nothing, a one percent move. In the early 1920s, ten points meant a ten percent move, quite a bit more important. For ease of charting, however, we suggest using

semilog scale only for long term plots, where the difference is especially noticeable. Arithmetic scale is quite acceptable for tracking hourly waves since a 300 point rally with the DJIA at 5000 is not much different in percentage terms from a 300 point rally with the DJIA at 6000. Thus, channeling techniques work acceptably well on arithmetic scale with shorter term moves.

6.2 Channeling Technique

Elliott noted that parallel trend channels typically mark the upper and lower boundaries of impulse waves, often with dramatic precision. The analyst should draw them in advance to assist in determining wave targets and provide clues to the future development of trends.

The initial channeling technique for an impulse requires at least three reference points. When wave three ends, connect the points labeled "1" and "3," then draw a parallel line touching the point labeled "2," as shown in Figure 2-8. This construction provides an estimated boundary for wave four. (In most cases, third waves travel far enough that the starting point is excluded from the final channel's touch points.)

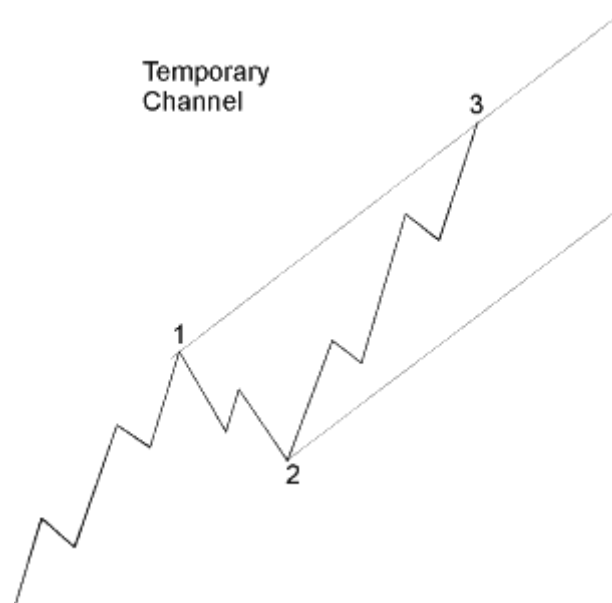


Figure 2-8

If the fourth wave ends at a point not touching the parallel, you must reconstruct the channel in order to estimate the boundary for wave five. First connect the ends of waves two and four. If waves one and three are normal, the upper parallel most accurately forecasts the end of wave five when drawn touching the peak of wave three, as in Figure 2-9. If wave three is abnormally strong, almost vertical, then a parallel drawn from its top may be too high. Experience has shown that a parallel to the baseline that touches the top of wave one is then more useful, as in the illustration of the rise in the price of gold bullion from August 1976 to March 1977 (see Figure 6-12). In some cases, it may be useful to draw both potential upper boundary lines to alert you to be especially attentive to the wave count and volume characteristics at those levels and then take appropriate action as the wave count warrants.

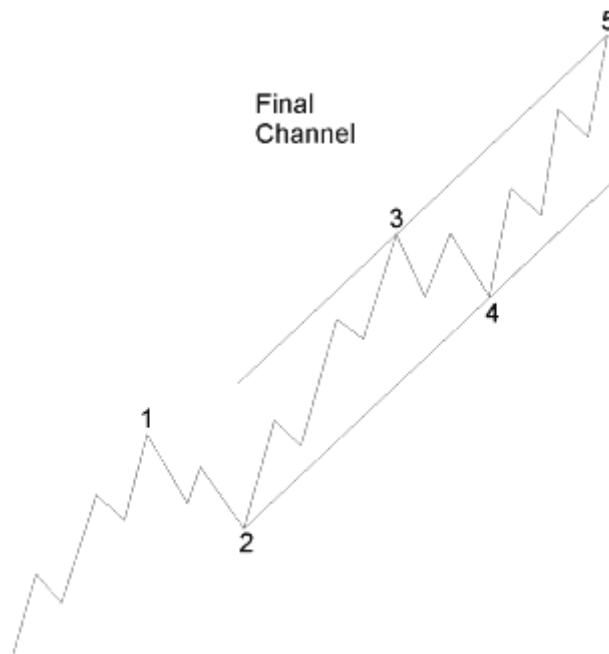


Figure 2-9

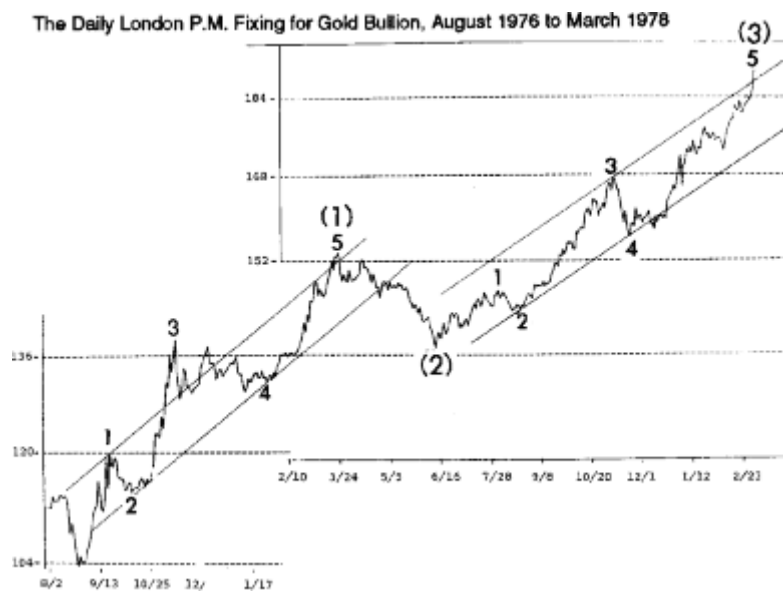


Figure 6-12

Throw-over

Within parallel channels and the converging lines of diagonal triangles, if a fifth wave approaches its upper trendline on declining volume, it is an indication that the end of the wave will meet or fall short of it. If volume is heavy as the fifth wave approaches its upper trendline, it indicates a possible penetration of the upper line, which Elliott called "throw-over." Near the point of throw-over, a fourth wave of small degree may trend sideways immediately below the parallel, allowing the fifth then to break it in a final gust of volume.

Throw-overs are occasionally telegraphed by a preceding "throw-under," either by wave 4 or by wave two of 5, as suggested by the drawing shown as Figure 2-10, from Elliott's book, *The Wave Principle*. They are confirmed by an immediate reversal back below the line. Throw-overs also occur, with the

same characteristics, in declining markets. Elliott correctly warned that throw-overs at large degrees cause difficulty in identifying the waves of smaller degree during the throw-over, as smaller degree channels are sometimes penetrated on the upside by the final fifth wave. Examples of throw-overs shown earlier in this course can be found in Figures 1-17 and 1-19.

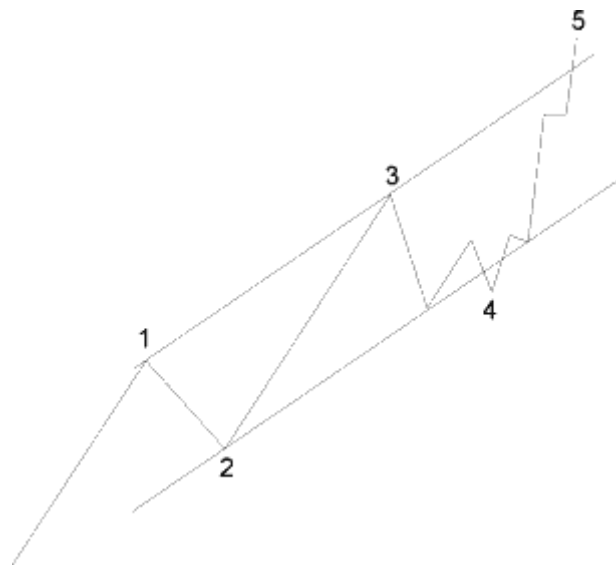


Figure 2-10

6.3 More Guidelines

Scale

The larger the degree, the more necessary a semilog scale usually becomes. On the other hand, the virtually perfect channels that were formed by the 1921-1929 market on semilog scale (see Figure 2-11) and the 1932-1937 market on arithmetic scale (see Figure 2-12) indicate that waves of the same degree will form the correct Elliott trend channel only when plotted selectively on the appropriate scale. On arithmetic scale, the 1920s bull market accelerates beyond the upper boundary, while on semilog scale the 1930s bull market falls far short of the upper boundary. Aside from this difference in channeling, these two waves of Cycle dimension are surprisingly similar: they create nearly the same multiples in price (six times and five times respectively), they both contain extended fifth waves, and the peak of the third wave is the same percentage gain above the bottom in each case. The essential difference between the two bull markets is the shape and time length of each individual subwave.

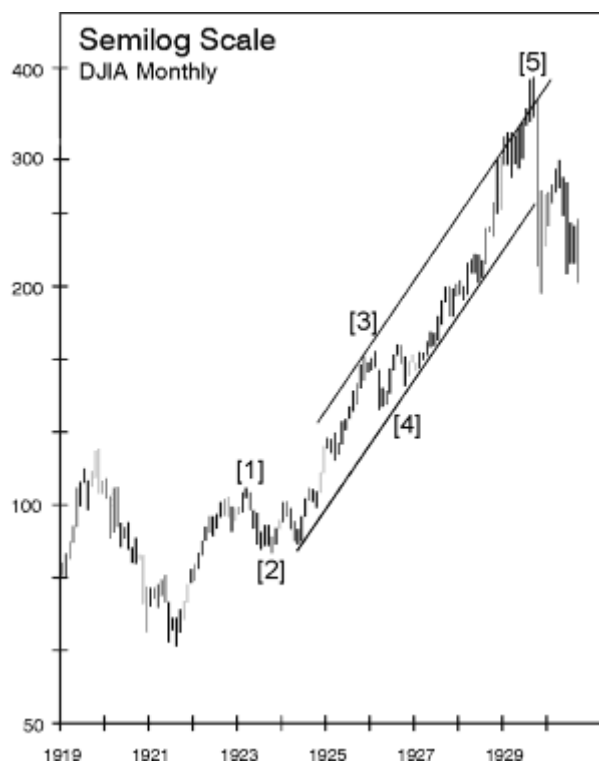


Figure 2-11

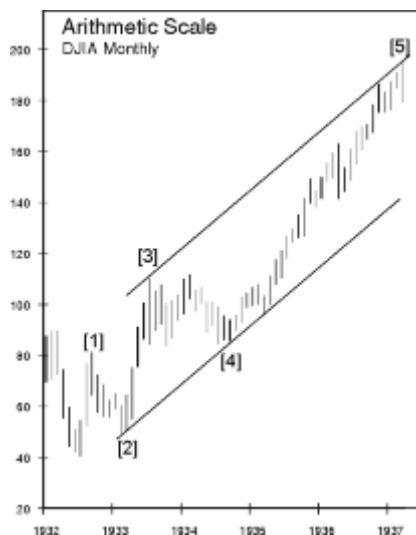


Figure 2-12

At most, we can state that the necessity for semilog scale indicates a wave that is in the process of acceleration, for whatever mass psychological reasons. Given a single price objective and a specific length of time allotted, anyone can draw a satisfactory hypothetical Elliott Wave channel from the same point of origin on both arithmetic and semilog scale by adjusting the slope of the waves to fit. Thus, the question of whether to expect a parallel channel on arithmetic or semilog scale is still unresolved as far as developing a definite tenet on the subject. If the price development at any point does not fall neatly within two parallel lines on the scale (either arithmetic or semilog) you are using, switch to the other scale in order to observe the channel in correct perspective. To stay on top of all developments, the analyst should always use both.

6.4 Volume

Elliott used volume as a tool for verifying wave counts and in projecting extensions. He recognized that in any bull market, volume has a natural tendency to expand and contract with the speed of price change. Late in a corrective phase, a decline in volume often indicates a decline in selling pressure. A low point in volume often coincides with a turning point in the market. In normal fifth waves below Primary degree, volume tends to be less than in third waves. If volume in an advancing fifth wave of less than Primary degree is equal to or greater than that in the third wave, an extension of the fifth is in force. While this outcome is often to be expected anyway if the first and third waves are about equal in length, it is an excellent warning of those rare times when both a third *and* a fifth wave are extended.

At Primary degree and greater, volume tends to be higher in an advancing fifth wave merely because of the natural long term growth in the number of participants in bull markets. Elliott noted, in fact, that volume at the terminal point of a bull market above Primary degree tends to run at an all-time high. Finally, as discussed earlier, volume often spikes briefly at points of throw-over at the peak of fifth waves, whether at a trend channel line or the terminus of a diagonal triangle. (Upon occasion, such points can occur simultaneously, as when a diagonal triangle fifth wave terminates right at the upper parallel of the channel containing the price action of one larger degree.) In addition to these few valuable observations, we have expanded upon the importance of volume in various sections of this course.

The "Right Look"

The overall appearance of a wave must conform to the appropriate illustration. Although any five-wave sequence can be forced into a three-wave count by labeling the first three subdivisions as one wave "A" as shown in Figure 2-13, it is incorrect to do so. The Elliott system would break down if such contortions were allowed. A long wave three with the end of wave four terminating well above the top of wave one must be classified as a five-wave sequence. Since wave A in this hypothetical case is composed of three waves, wave B would be expected to drop to about the start of wave A, as in a flat correction, which it clearly does not. While the internal count of a wave is a guide to its classification, the right overall shape is, in turn, often a guide to its correct internal count.

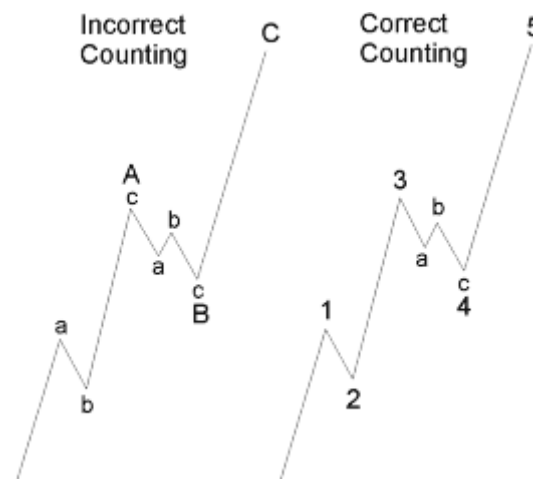


Figure 2-13

The "right look" of a wave is dictated by all the considerations we have outlined so far in the first two chapters. In our experience, we have found it extremely dangerous to allow our emotional involvement with the market to let us accept wave counts that reflect disproportionate wave relationships or misshapen patterns merely on the basis that the Wave Principle's patterns are somewhat elastic.

Lesson 7

7.1 Wave Personality

The idea of wave personality is a substantial expansion of the Wave Principle. It has the advantages of bringing human behavior more personally into the equation and even more important, of enhancing the utility of standard technical analysis.

The personality of each wave in the Elliott sequence is an integral part of the reflection of the mass psychology it embodies. The progression of mass emotions from pessimism to optimism and back again tends to follow a similar path each time around, producing similar circumstances at corresponding points in the wave structure. The personality of each wave type is usually manifest whether the wave is of Grand Supercycle degree or Subminuette. These properties not only forewarn the analyst about what to expect in the next sequence but at times can help determine one's present location in the progression of waves, when for other reasons the count is unclear or open to differing interpretations. As waves are in the process of unfolding, there are times when several different wave counts are perfectly admissible under all known Elliott rules. It is at these junctures that a knowledge of wave personality can be invaluable. If the analyst recognizes the character of a single wave, he can often correctly interpret the complexities of the larger pattern. The following discussions relate to an underlying bull market picture, as illustrated in Figures 2-14 and 2-15. These observations apply in reverse when the actionary waves are downward and the reactionary waves are upward.

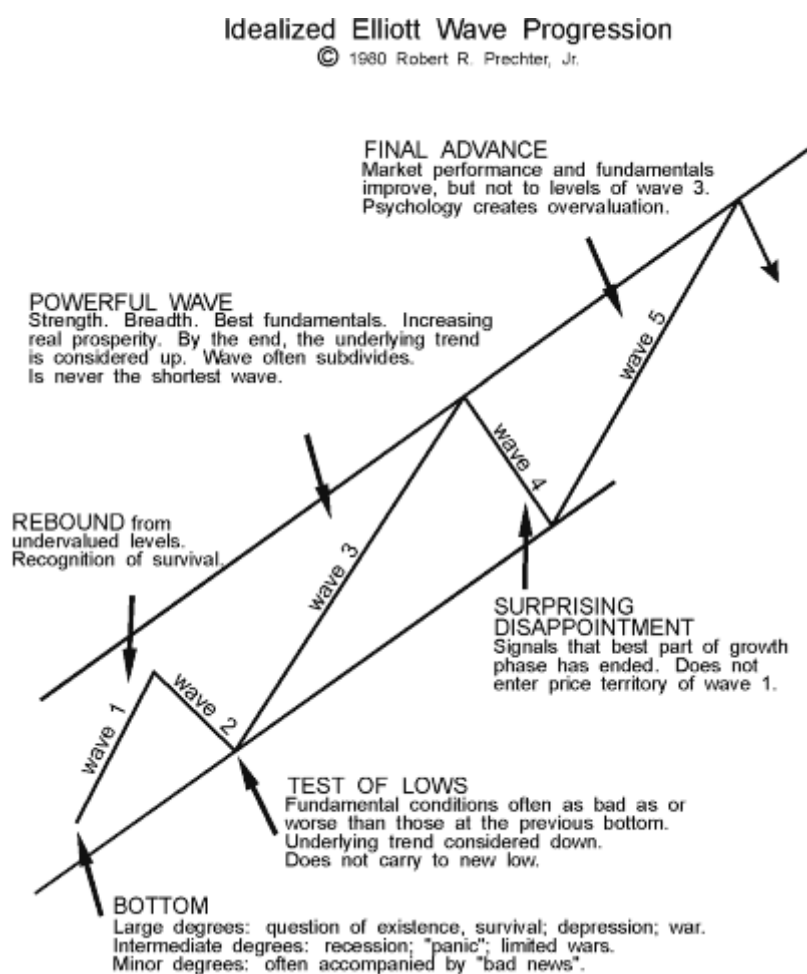


Figure 2-14

7.2 Wave Personality

1) **First** waves — As a rough estimate, about half of first waves are part of the "basing" process and thus tend to be heavily corrected by wave two. In contrast to the bear market rallies within the previous decline, however, this first wave rise is technically more constructive, often displaying a subtle increase in volume and breadth. Plenty of short selling is in evidence as the majority has finally become convinced that the overall trend is down. Investors have finally gotten "one more rally to sell on," and they take advantage of it. The other fifty percent of first waves rise from either large bases formed by the previous correction, as in 1949, from downside failures, as in 1962, or from extreme compression, as in both 1962 and 1974. From such beginnings, first waves are dynamic and only moderately retraced.

2) **Second** waves — Second waves often retrace so much of wave one that most of the advancement up to that time is eroded away by the time it ends. This is especially true of call option purchases, as premiums sink drastically in the environment of fear during second waves. At this point, investors are thoroughly convinced that the bear market is back to stay. Second waves often produce downside non-confirmations and Dow Theory "buy spots," when low volume and volatility indicate a drying up of selling pressure.

3) **Third** waves — Third waves are wonders to behold. They are strong and broad, and the trend at this point is unmistakable. Increasingly favorable fundamentals enter the picture as confidence returns. Third waves usually generate the greatest volume and price movement and are most often the extended wave in a series. It follows, of course, that the third wave of a third wave, and so on, will be the most volatile point of strength in any wave sequence. Such points invariably produce breakouts, "continuation" gaps, volume expansions, exceptional breadth, major Dow Theory trend confirmations and runaway price movement, creating large hourly, daily, weekly, monthly or yearly gains in the market, depending on the degree of the wave. Virtually all stocks participate in third waves. Besides the personality of "B" waves, that of third waves produces the most valuable clues to the wave count as it unfolds.

4) **Fourth** waves — Fourth waves are predictable in both depth (see Lesson 11) and form, because by alternation they should differ from the previous second wave of the same degree.

More often than not they trend sideways, building the base for the final fifth wave move. Lagging stocks build their tops and begin declining during this wave, since only the strength of a third wave was able to generate any motion in them in the first place. This initial deterioration in the market sets the stage for non-confirmations and subtle signs of weakness during the fifth wave.

5) **Fifth** waves — Fifth waves in stocks are always less dynamic than third waves in terms of breadth. They usually display a slower maximum speed of price change as well, although if a fifth wave is an extension, speed of price change in the third of the fifth can exceed that of the third wave. Similarly, while it is common for volume to increase through successive impulse waves at Cycle degree or larger, it usually happens below Primary degree only if the fifth wave extends. Otherwise, look for *lesser* volume as a rule in a fifth wave as opposed to the third. Market dabblers sometimes call for "blowoffs" at the end of long trends, but the stock market has no history of reaching maximum acceleration at a peak. Even if a fifth wave extends, the fifth of the fifth will lack the dynamism of what preceded it. During fifth advancing waves, optimism runs extremely high, despite a narrowing of breadth. Nevertheless, market action does improve relative to prior corrective wave rallies. For example, the year-end rally in 1976 was unexciting in the Dow, but it was nevertheless a motive wave as opposed to the preceding corrective wave advances in April, July and September, which, by contrast, had even less influence on the secondary indexes and the cumulative advance-decline line. As a monument to the optimism that fifth waves can produce, the market forecasting services polled two weeks after the conclusion of that rally turned in the lowest percentage of "bears," 4.5%, in the history of the recorded figures *despite* that fifth wave's failure to make a new high!

7.3 Ideal Wave Personality

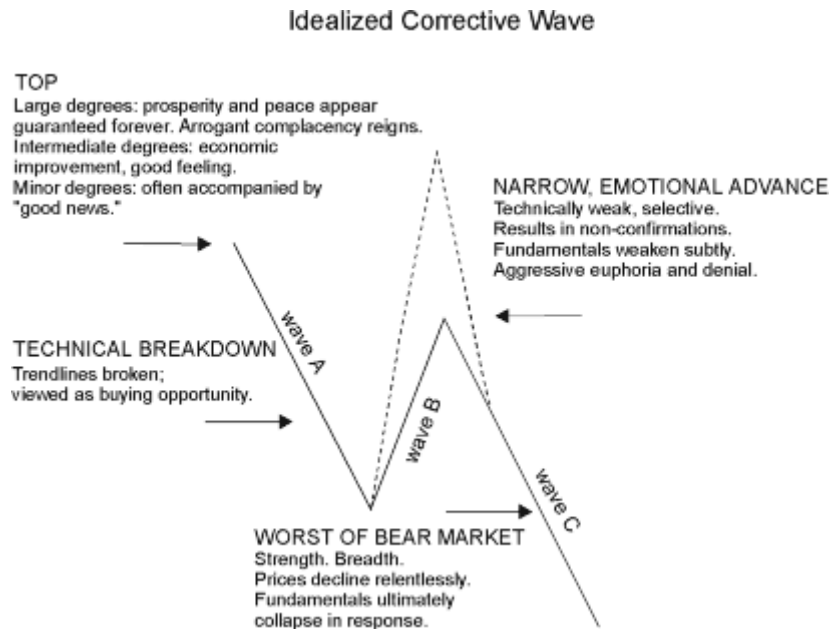


Figure 2-15

6) **"A" waves** — During "A" waves of bear markets, the investment world is generally convinced that this reaction is just a pullback pursuant to the next leg of advance. The public surges to the buy side despite the first really technically damaging cracks in individual stock patterns. The "A" wave sets the tone for the "B" wave to follow. A five-wave A indicates a zigzag for wave B, while a three-wave A indicates a flat or triangle.

7) **"B" waves** — "B" waves are phonies. They are sucker plays, bull traps, speculators' paradise, orgies of odd-lotter mentality or expressions of dumb institutional complacency (or both). They often involve a focus on a narrow list of stocks, are often "unconfirmed" (Dow Theory is covered in Lesson 28) by other averages, are rarely technically strong, and are virtually always doomed to complete retracement by wave C. If the analyst can easily say to himself, "There is something wrong with this market," chances are it's a "B" wave. "X" waves and "D" waves in expanding triangles, both of which are corrective wave advances, have the same characteristics. Several examples will suffice to illustrate the point.

— The upward correction of 1930 was wave B within the 1929-1932 A-B-C zigzag decline. Robert Rhea describes the emotional climate well in his opus, *The Story of the Averages* (1934):

...many observers took it to be a bull market signal. I can remember having shorted stocks early in December, 1929, after having completed a satisfactory short position in October. When the slow but steady advance of January and February carried above [the previous high], I became panicky and covered at considerable loss. ...I forgot that the rally might normally be expected to retrace possibly 66 percent or more of the 1929 downswing. Nearly everyone was proclaiming a new bull market. Services were extremely bullish, and the upside volume was running higher than at the peak in 1929.

— The 1961-1962 rise was wave (b) in an (a)-(b)-(c) expanded flat correction. At the top in early 1962, stocks were selling at unheard of price/earnings multiples that had not been seen up to that time and have not been seen since. Cumulative breadth had already peaked along with the top of the third wave in 1959.

— The rise from 1966 to 1968 was wave [B]* in a corrective pattern of Cycle degree. Emotionalism had gripped the public and "cheapies" were skyrocketing in the speculative fever, unlike the orderly and usually fundamentally justifiable participation of the secondaries within first and third waves. The Dow Industrials struggled unconvincingly higher throughout the advance and finally refused to confirm the phenomenal new highs in the secondary indexes.

— In 1977, the Dow Jones Transportation Average climbed to new highs in a "B" wave, miserably unconfirmed by the Industrials. Airlines and truckers were sluggish. Only the coal-carrying rails were participating as part of the energy play. Thus, breadth within the index was conspicuously lacking, confirming again that good breadth is generally a property of impulse waves, not corrections.

As a general observation, "B" waves of Intermediate degree and lower usually show a diminution of volume, while "B" waves of Primary degree and greater can display volume heavier than that which accompanied the preceding bull market, usually indicating wide public participation.

8) **"C" waves** — Declining "C" waves are usually devastating in their destruction. They are third waves and have most of the properties of third waves. It is during this decline that there is virtually no place to hide except cash. The illusions held throughout waves A and B tend to evaporate and fear takes over. "C" waves are persistent and broad. 1930-1932 was a "C" wave. 1962 was a "C" wave. 1969-1970 and 1973-1974 can be classified as "C" waves. Advancing "C" waves within upward corrections in larger bear markets are just as dynamic and can be mistaken for the start of a new upswing, especially since they unfold in five waves. The October 1973 rally (see Figure 1-37), for instance, was a "C" wave in an inverted expanded flat correction.

9) **"D" waves** — "D" waves in all but expanding triangles are often accompanied by increased volume. This is true probably because "D" waves in non-expanding triangles are hybrids, part corrective, yet having some characteristics of first waves since they follow "C" waves and are not fully retraced. "D" waves, being advances within corrective waves, are as phony as "B" waves. The rise from 1970 to 1973 was wave [D] within the large wave IV of Cycle degree. The "one-decision" complacency that characterized the attitude of the average institutional fund manager at the time is well documented. The area of participation again was narrow, this time the "nifty fifty" growth and glamour issues. Breadth, as well as the Transportation Average, topped early, in 1972, and refused to confirm the extremely high multiples bestowed upon the favorite fifty. Washington was inflating at full steam to sustain the illusory prosperity during the entire advance in preparation for the election. As with the preceding wave [B], "phony" was an apt description.

10) **"E" waves** — "E" waves in triangles appear to most market observers to be the dramatic kickoff of a new downtrend after a top has been built. They almost always are accompanied by strongly supportive news. That, in conjunction with the tendency of "E" waves to stage a false breakdown through the triangle boundary line, intensifies the bearish conviction of market participants at precisely the time that they should be preparing for a substantial move in the opposite direction. Thus, "E" waves, being ending waves, are attended by a psychology as emotional as that of fifth waves.

7.4 Wave Tendencies

Because the tendencies discussed here are not inevitable, they are stated not as rules, but as guidelines. Their lack of inevitability nevertheless detracts little from their utility. For example, take a look at Figure 2-16, an hourly chart showing the first four Minor waves in the DJIA rally off the March 1, 1978 low. The waves are textbook Elliott from beginning to end, from the length of waves to the volume pattern (not shown) to the trend channels to the guideline of equality to the retracement by the "a" wave following the extension to the expected low for the fourth wave to the perfect internal counts to alternation to the Fibonacci time sequences to the Fibonacci ratio relationships embodied within. It might be worth noting that 914 would be a reasonable target in that it would mark a .618 retracement of the 1976-1978 decline.



Figure 2-16 (Click Image To Enlarge)

There are exceptions to guidelines, but without those, market analysis would be a science of exactitude, not one of probability. Nevertheless, with a thorough knowledge of the guide lines of wave structure, you can be quite confident of your wave count. In effect, you can use the market action to confirm the wave count as well as use the wave count to predict market action.

Notice also that Elliott Wave guidelines cover most aspects of traditional technical analysis, such as market momentum and investor sentiment. The result is that traditional technical analysis now has a greatly increased value in that it serves to aid the identification of the market's exact position in the Elliott Wave structure. To that end, using such tools is by all means encouraged.

7.5 Learning the Basics

With a knowledge of the tools in Lessons 1 through 15, any dedicated student can perform expert Elliott Wave analysis. People who neglect to study the subject thoroughly or to apply the tools rigorously have given up before really trying. The best learning procedure is to keep an hourly chart and try to fit all the wiggles into Elliott Wave patterns, while keeping an open mind for all the possibilities. Slowly the scales should drop from your eyes, and you will continually be amazed at what you see.

It is important to remember that while investment tactics always must go with the most valid wave count, knowledge of alternative possibilities can be extremely helpful in adjusting to unexpected events, putting them immediately into perspective, and adapting to the changing market framework. While the rigidities of the rules of wave formation are of great value in choosing entry and exit points, the flexibilities in the admissible patterns eliminate cries that whatever the market is doing now is "impossible."

"When you have eliminated the impossible, whatever remains, *however improbable*, must be the truth." Thus eloquently spoke Sherlock Holmes to his constant companion, Dr. Watson, in Arthur Conan Doyle's *The Sign of Four*. This one sentence is a capsule summary of what one needs to know to be successful with Elliott. The best approach is deductive reasoning. By knowing what Elliott rules will not allow, one can deduce that whatever remains must be the most likely course for the market. Applying all the rules of extensions, alternation, overlapping, channeling, volume and the rest, the analyst has a much more formidable arsenal than one might imagine at first glance. Unfortunately for many, the approach requires thought and work and rarely provides a mechanical signal. However, this kind of thinking, basically an elimination process, squeezes the best out of what Elliott has to offer and besides, it's fun!

As an example of such deductive reasoning, take another look at Figure 1-14, reproduced below:



Figure 1-14

Cover up the price action from November 17, 1976 forward. Without the wave labels and boundary lines, the market would appear as formless. But with the Wave Principle as a guide, the meaning of the structures becomes clear. Now ask yourself, how would you go about predicting the next movement? Here is Robert Prechter's analysis from that date, from a personal letter to A.J. Frost, summarizing a report he issued for Merrill Lynch the previous day:

Enclosed you will find my current opinion outlined on a recent Trendline chart, although I use only hourly point charts to arrive at these conclusions. My argument is that the third Primary wave, begun in October of 1975, *has not completed* its course as yet, and that the fifth Intermediate wave of that Primary is now underway. First and most important, I am convinced that October 1975 to March 1976 was so far a three-wave affair, not a five, and that only the possibility of a failure on May 11th could complete that wave as a five. However, the construction *following* that possible "failure" does not satisfy me as correct, since the first downleg to 956.45 would be of five waves and the entire ensuing construction is obviously a flat. Therefore, I think that we have been in a fourth corrective wave since March 24th. This corrective wave satisfies *completely* the requirements for an expanding triangle formation, which of course can only be a fourth wave. The trendlines concerned are uncannily accurate, as is the downside objective, obtained by multiplying the first important length of decline (March 24th to June 7th, 55.51 points) by 1.618 to obtain 89.82 points. 89.82 points from the orthodox high of the third Intermediate wave at 1011.96 gives a downside target of 922, which was hit last week (actual hourly low 920.62) on November 11th. This would suggest now a fifth Intermediate back to new highs, completing the third Primary wave. The only problem I can see with this interpretation is that Elliott suggests that fourth wave declines usually hold above the previous fourth wave decline of lesser degree, in this case 950.57 on February 17th, which of course has been broken on the downside. I have found, however, that this rule is not steadfast. The reverse symmetrical triangle formation should be followed by a rally only approximating the width of the widest part of the triangle. Such a rally would suggest 1020-1030 and fall far short of the trendline target of 1090-1100. Also, *within* third waves, the first and fifth subwaves tend toward equality in time and magnitude. Since the first wave (Oct. 75-Dec.75) was a 10% move in two months, this fifth should cover about 100 points (1020-1030) and peak in January 1977, again short of the trendline mark.

Now uncover the rest of the chart to see how all these guidelines helped in assessing the market's likely path.

Christopher Morley once said, "Dancing is a wonderful training for girls. It is the first way they learn to guess what a man is going to do before he does it." In the same way, the Wave Principle trains the analyst to discern what the market is likely to do before it does it.

After you have acquired an Elliott "touch," it will be forever with you, just as a child who learns to ride a bicycle never forgets. At that point, catching a turn becomes a fairly common experience and not really too difficult. Most important, in giving you a feeling of confidence as to where you are in the progress of the market, a knowledge of Elliott can prepare you psychologically for the inevitable fluctuating nature of price movement and free you from sharing the widely practiced analytical error of forever projecting today's trends linearly into the future.

7.6 Practical Application

The Wave Principle is unparalleled in providing an overall perspective on the position of the market most of the time. Most important to individuals, portfolio managers and investment corporations is that the Wave Principle often indicates in advance the relative *magnitude* of the next period of market progress or regress. Living in harmony with those trends can make the difference between success and failure in financial affairs.

Despite the fact that many analysts do not treat it as such, the Wave Principle is by all means an objective study, or as Collins put it, "a disciplined form of technical analysis." Bolton used to say that one of the hardest things he had to learn was to believe what he saw. If the analyst does not believe what he sees, he is likely to read into his analysis what he thinks should be there for some other reason. At this point, his count becomes subjective. Subjective analysis is dangerous and destroys the value of any market approach.

What the Wave Principle provides is an objective means of assessing the relative *probabilities* of possible future paths for the market. At any time, two or more valid wave interpretations are usually acceptable by the *rules* of the Wave Principle. The rules are highly specific and keep the number of valid alternatives to a minimum. Among the valid alternatives, the analyst will generally regard as preferred the interpretation that satisfies the largest number of guidelines, and so on. As a result, competent analysts applying the rules and guidelines of the Wave Principle objectively should usually agree on the *order* of probabilities for various possible outcomes at any particular time. That order can usually be stated with certainty. Let no one assume, however, that certainty about the order of probabilities is the same as certainty about one specific outcome. Under only the rarest of circumstances does the analyst ever *know exactly* what the market is going to do. One must understand and accept that even an approach that can identify high odds for a fairly specific outcome will be wrong some of the time. Of course, such a result is a far better performance than any other approach to market forecasting provides.

Using Elliott, it is often possible to make money even when you are in error. For instance, after a minor low that you erroneously consider of major importance, you may recognize *at a higher level* that the market is vulnerable again to new lows. A clear-cut *three-wave* rally following the minor low rather than the necessary five gives the signal, since a three-wave rally is the sign of an upward correction. Thus, what happens *after* the turning point often helps confirm or refute the assumed status of the low or high, well in advance of danger.

Even if the market allows no such graceful exit, the Wave Principle still offers exceptional value. Most other approaches to market analysis, whether fundamental, technical or cyclical, have no good way of forcing a change of opinion if you are wrong. The Wave Principle, in contrast, provides a built-in objective method for changing your mind. Since Elliott Wave analysis is based upon price patterns, a pattern identified as having been completed is either over or it isn't. If the market changes direction, the analyst has caught the turn. If the market moves beyond what the apparently completed pattern allows, the conclusion is wrong, and any funds at risk can be reclaimed immediately. Investors using the Wave Principle can prepare themselves psychologically for such outcomes through the continual updating of the *second best interpretation*, sometimes called the "alternate count." Because applying the Wave Principle is an exercise in probability, the ongoing maintenance of alternative wave counts is an essential part of investing with it. In the event that the market violates the expected scenario, the alternate count immediately becomes the investor's new preferred count. If you're thrown by your horse, it's useful to land right atop another.

Of course, there are often times when, despite a rigorous analysis, the question may arise as to how a developing move is to be counted, or perhaps classified as to degree. When there is no clearly preferred interpretation, the analyst must wait until the count resolves itself, in other words, to "sweep it under the rug until the air clears," as Bolton suggested. Almost always, subsequent moves will clarify the status of previous waves by revealing their position in the pattern of the next higher degree. When subsequent waves clarify the picture, the probability that a turning point is at hand can suddenly and excitingly rise to nearly 100%.

7.7 Practical Application

The ability to *identify* junctures is remarkable enough, but the Wave Principle is the only method of analysis which also provides guidelines for *forecasting*, as outlined in Lessons 10 through 15 and 20 through 25 of this course. Many of these guidelines are specific and can occasionally yield results of stunning precision. If indeed markets are patterned, and if those patterns have a recognizable geometry, then regardless of the variations allowed, certain price and time relationships are likely to recur. In fact, real world experience shows that they do.

It is our practice to try to determine in advance where the next move will likely take the market. One advantage of setting a target is that it gives a sort of backdrop against which to monitor the market's actual path. This way, you are alerted quickly when something is wrong and can shift your interpretation to a more appropriate one if the market does not do what is expected. If you then learn the reasons for your mistakes, the market will be less likely to mislead you in the future.

Still, no matter what your convictions, it pays never to take your eye off what is happening in the wave structure in real time. Although prediction of target levels well in advance can be done surprisingly often, such predictions are not required in order to make money in the stock market. Ultimately, the market is the message, and a change in behavior can dictate a change in outlook. All one really needs to know *at the time* is whether to be bullish, bearish or neutral, a decision that can sometimes be made with a swift glance at a chart.

Of the many approaches to stock market analysis, the Elliott Wave Principle, in our view, offers the best tool for identifying market turns as they are approached. If you keep an hourly chart, the fifth of the fifth of the fifth in a primary trend alerts you within hours of a major change in direction by the market. It is a thrilling experience to pinpoint a turn, and the Wave Principle is the only approach that can occasionally provide the opportunity to do so. Elliott may not be the perfect formulation since the stock market is part of life and no formula can enclose it or express it completely. However, the Wave Principle is without a doubt the single most comprehensive approach to market analysis and, viewed in its proper light, delivers everything it promises.

Lesson 8

8.1 Introducing Fibonacci



***Statue of Leonardo Fibonacci, Pisa, Italy.
The inscription reads, "A. Leonardo Fibonacci, Insigne
Matematico Pisano del Secolo XII."
Photo by Robert R. Prechter, Sr.***

HISTORICAL AND MATHEMATICAL BACKGROUND OF THE WAVE PRINCIPLE

The Fibonacci (pronounced fib-eh-nah'-chee) sequence of numbers was discovered (actually rediscovered) by Leonardo Fibonacci da Pisa, a thirteenth century mathematician. We will outline the historical background of this amazing man and then discuss more fully the sequence (technically it is a sequence and not a series) of numbers that bears his name. When Elliott wrote *Nature's Law*, he referred specifically to the Fibonacci sequence as the mathematical basis for the Wave Principle. It is sufficient to state at this point that the stock market has a propensity to demonstrate a form that can be aligned with the form present in the Fibonacci sequence. (For a further discussion of the mathematics behind the Wave Principle, see "Mathematical Basis of Wave Theory," by Walter E. White, in New Classics Library's forthcoming book.)

In the early 1200s, Leonardo Fibonacci of Pisa, Italy published his famous *Liber Abacci* (Book of Calculation) which introduced to Europe one of the greatest mathematical discoveries of all time, namely the decimal system, including the positioning of zero as the first digit in the notation of the

number scale. This system, which included the familiar symbols 0, 1, 2, 3, 4, 5, 6, 7, 8 and 9, became known as the Hindu-Arabic system, which is now universally used.

Under a true digital or place-value system, the actual value represented by any symbol placed in a row along with other symbols depends not only on its basic numerical value but also on its position in the row, i.e., 58 has a different value from 85. Though thousands of years earlier the Babylonians and Mayas of Central America separately had developed digital or place-value systems of numeration, their methods were awkward in other respects. For this reason, the Babylonian system, which had been the first to use zero and place values, was never carried forward into the mathematical systems of Greece, or even Rome, whose numeration comprised the seven symbols I, V, X, L, C, D, and M, with non-digital values assigned to those symbols. Addition, subtraction, multiplication and division in a system using these non-digital symbols is not an easy task, especially when large numbers are involved. Paradoxically, to overcome this problem, the Romans used the very ancient digital device known as the abacus. Because this instrument is digitally based and contains the zero principle, it functioned as a necessary supplement to the Roman computational system. Throughout the ages, bookkeepers and merchants depended on it to assist them in the mechanics of their tasks. Fibonacci, after expressing the basic principle of the abacus in *Liber Abacci*, started to use his new system during his travels. Through his efforts, the new system, with its easy method of calculation, was eventually transmitted to Europe. Gradually the old usage of Roman numerals was replaced with the Arabic numeral system. The introduction of the new system to Europe was the first important achievement in the field of mathematics since the fall of Rome over seven hundred years before. Fibonacci not only kept mathematics alive during the Middle Ages, but laid the foundation for great developments in the field of higher mathematics and the related fields of physics, astronomy and engineering.

8.2 Introducing Fibonacci

Although the world later almost lost sight of Fibonacci, he was unquestionably a man of his time. His fame was such that Frederick II, a scientist and scholar in his own right, sought him out by arranging a visit to Pisa. Frederick II was Emperor of the Holy Roman Empire, the King of Sicily and Jerusalem, scion of two of the noblest families in Europe and Sicily, and the most powerful prince of his day. His ideas were those of an absolute monarch, and he surrounded himself with all the pomp of a Roman emperor.

The meeting between Fibonacci and Frederick II took place in 1225 A.D. and was an event of great importance to the town of Pisa. The Emperor rode at the head of a long procession of trumpeters, courtiers, knights, officials and a menagerie of animals. Some of the problems the Emperor placed before the famous mathematician are detailed in *Liber Abacci*. Fibonacci apparently solved the problems posed by the Emperor and forever more was welcome at the King's Court. When Fibonacci revised *Liber Abacci* in 1228 A.D., he dedicated the revised edition to Frederick II.

It is almost an understatement to say that Leonardo Fibonacci was the greatest mathematician of the Middle Ages. In all, he wrote three major mathematical works: the *Liber Abacci*, published in 1202 and revised in 1228, *Practica Geometriae*, published in 1220, and *Liber Quadratorum*. The admiring citizens of Pisa documented in 1240 A.D. that he was "a discreet and learned man," and very recently Joseph Gies, a senior editor of the Encyclopedia Britannica, stated that future scholars will in time "give Leonard of Pisa his due as one of the world's great intellectual pioneers." His works, after all these years, are only now being translated from Latin into English. For those interested, the book entitled *Leonard of Pisa and the New Mathematics of the Middle Ages*, by Joseph and Frances Gies, is an excellent treatise on the age of Fibonacci and his works.

Although he was the greatest mathematician of medieval times, Fibonacci's only monuments are a statue across the Arno River from the Leaning Tower and two streets which bear his name, one in Pisa and the other in Florence. It seems strange that so few visitors to the 179-foot marble Tower of Pisa have ever heard of Fibonacci or seen his statue. Fibonacci was a contemporary of Bonanno, the architect of the Tower, who started building in 1174 A.D. Both men made contributions to the world, but the one whose influence far exceeds the other's is almost unknown.

8.3 The Fibonacci Sequence

In *Liber Abacci*, a problem is posed that gives rise to the sequence of numbers 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, and so on to infinity, known today as the Fibonacci sequence. The problem is this:

How many pairs of rabbits placed in an enclosed area can be produced in a single year from one pair of rabbits if each pair gives birth to a new pair each month starting with the second month?

In arriving at the solution, we find that each pair, including the first pair, needs a month's time to mature, but once in production, begets a new pair each month. The number of pairs is the same at the beginning of each of the first two months, so the sequence is 1, 1. This first pair finally doubles its number during the second month, so that there are two pairs at the beginning of the third month. Of these, the older pair begets a third pair the following month so that at the beginning of the fourth month, the sequence expands 1, 1, 2, 3. Of these three, the two older pairs reproduce, but not the youngest pair, so the number of rabbit pairs expands to five. The next month, three pairs reproduce so the sequence expands to 1, 1, 2, 3, 5, 8 and so forth. Figure 3-1 shows the Rabbit Family Tree with the family growing with logarithmic acceleration. Continue the sequence for a few years and the numbers become astronomical. In 100 months, for instance, we would have to contend with 354,224,848,179,261,915,075 pairs of rabbits. The Fibonacci sequence resulting from the rabbit problem has many interesting properties and reflects an almost constant relationship among its components.

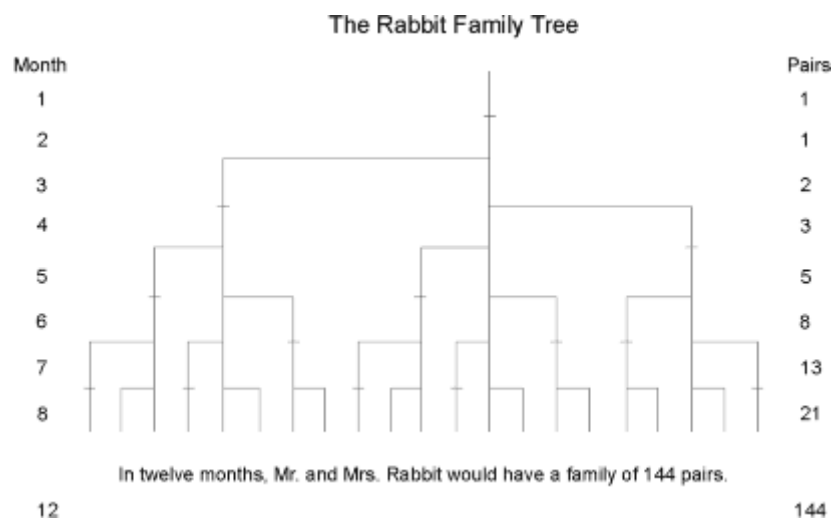


Figure 3-1

The sum of any two adjacent numbers in the sequence forms the next higher number in the sequence, viz., 1 plus 1 equals 2, 1 plus 2 equals 3, 2 plus 3 equals 5, 3 plus 5 equals 8, and so on to infinity.

The Golden Ratio

After the first several numbers in the sequence, the ratio of any number to the next higher is approximately .618 to 1 and to the next lower number approximately 1.618 to 1. The further along the sequence, the closer the ratio approaches *phi* (denoted ϕ) which is an irrational number, .618034.... Between alternate numbers in the sequence, the ratio is approximately .382, whose inverse is 2.618. Refer to Figure 3-2 for a ratio table interlocking all Fibonacci numbers from 1 to 144.

Fibonacci Ratio Table

		NUMERATOR											
		1	2	3	5	8	13	21	34	55	89	144	
DENOMINATOR	1	1.00	2.00	3.00	5.00	8.00	13.00	21.00	34.00	55.00	89.00	144.00	
	2	.50	1.00	1.50	2.50	4.00	6.50	10.50	17.00	27.50	44.50	72.00	
	3	.333	.667	1.00	1.667	2.667	4.33	7.00	11.33	18.33	29.67	48.00	
	5	.20	.40	.60	1.00	1.60	2.60	4.20	6.80	11.00	17.80	28.80	
	8	.125	.25	.375	.625	1.00	1.625	2.625	4.25	6.875	11.125	18.00	
	13	.077	.154	.231	.385	.615	1.00	1.615	2.615	4.23	6.846	11.077	
	21	.0476	.0952	.1429	.238	.381	.619	1.00	1.619	2.619	4.238	6.857	
	34	.0294	.0588	.0882	.147	.235	.3824	.6176	1.00	1.618	2.618	4.235	
	55	.01818	.03636	.0545	.0909	.1455	.236	.3816	.618	1.00	1.618	2.618	
	89	.011236	.02247	.0337	.05518	.08989	.140	.236	.382	.618	1.00	1.618	
	144	.006944	.013889	.0208	.0347	.05556	.0903	.1456	.236	.382	.618	1.00	

Toward perfect ratios

Figure 3-2

Phi is the only number that when added to 1 yields its inverse: $.618 + 1 = 1 \div .618$. This alliance of the additive and the multiplicative produces the following sequence of equations:

$$.618^2 = 1 - .618,$$

$$.618^3 = .618 - .618^2,$$

$$.618^4 = .618^2 - .618^3,$$

$$.618^5 = .618^3 - .618^4, \text{ etc.}$$

or alternatively,

$$1.618^2 = 1 + 1.618,$$

$$1.618^3 = 1.618 + 1.618^2,$$

$$1.618^4 = 1.618^2 + 1.618^3,$$

$$1.618^5 = 1.618^3 + 1.618^4, \text{ etc.}$$

Some statements of the interrelated properties of these four main ratios can be listed as follows:

1) $1.618 - .618 = 1,$

2) $1.618 \times .618 = 1,$

3) $1 - .618 = .382,$

4) $.618 \times .618 = .382,$

5) $2.618 - 1.618 = 1,$

6) $2.618 \times .382 = 1,$

7) $2.618 \times .618 = 1.618,$

8) $1.618 \times 1.618 = 2.618.$

Besides 1 and 2, any Fibonacci number multiplied by four, when added to a selected Fibonacci number, gives another Fibonacci number, so that:

$$3 \times 4 = 12; + 1 = 13,$$

$$5 \times 4 = 20; + 1 = 21,$$

$$8 \times 4 = 32; + 2 = 34,$$

$$13 \times 4 = 52; + 3 = 55,$$

$$21 \times 4 = 84; + 5 = 89, \text{ and so on.}$$

8.4 The Fibonacci Sequence

As the new sequence progresses, a third sequence begins in those numbers that are added to the 4x multiple. This relationship is possible because the ratio between *second* alternate Fibonacci numbers is 4.236, where .236 is both its inverse *and* its difference from the number 4. This continuous series-building property is reflected at other multiples for the same reasons.

1.618 (or .618) is known as the Golden Ratio or Golden Mean. Its proportions are pleasing to the eye and an important phenomenon in music, art, architecture and biology. William Hoffer, writing for the December 1975 *Smithsonian Magazine*, said:

...the proportion of .618034 to 1 is the mathematical basis for the shape of playing cards and the Parthenon, sunflowers and snail shells, Greek vases and the spiral galaxies of outer space. The Greeks based much of their art and architecture upon this proportion. They called it "the golden mean."

Fibonacci's abracadabric rabbits pop up in the most unexpected places. The numbers are unquestionably part of a mystical natural harmony that feels good, looks good and even sounds good. Music, for example, is based on the 8-note octave. On the piano this is represented by 8 white keys, 5 black ones — 13 in all. It is no accident that the musical harmony that seems to give the ear its greatest satisfaction is the major sixth. The note A vibrates at a ratio of .62500 to the note C. A mere .006966 away from the exact golden mean, the proportions of the major sixth set off good vibrations in the cochlea of the inner ear — an organ that just happens to be shaped in a logarithmic spiral.

The continual occurrence of Fibonacci numbers and the golden spiral in nature explains precisely why the proportion of .618034 to 1 is so pleasing in art. Man can see the image of life in art that is based on the golden mean.

Nature uses the Golden Ratio in its most intimate building blocks and in its most advanced patterns, in forms as minuscule as atomic structure, microtubules in the brain and DNA molecules to those as large as planetary orbits and galaxies. It is involved in such diverse phenomena as quasi crystal arrangements, planetary distances and periods, reflections of light beams on glass, the brain and nervous system, musical arrangement, and the structures of plants and animals. Science is rapidly demonstrating that there is indeed a basic proportional principle of nature. By the way, you are holding your mouse with your *five* appendages, all but one of which have *three* jointed parts, *five* digits at the end, and *three* jointed sections to each digit.

8.5 Fibonacci Geometry

The Golden Section

Any length can be divided in such a way that the ratio between the smaller part and the larger part is equivalent to the ratio between the larger part and the whole (see Figure 3-3). That ratio is always .618.



Figure 3-3

The Golden Section occurs throughout nature. In fact, the human body is a tapestry of Golden Sections (see Figure 3-9) in everything from outer dimensions to facial arrangement. "Plato, in his *Timaeus*," says Peter Tompkins, "went so far as to consider *phi*, and the resulting Golden Section proportion, the most binding of all mathematical relations, and considered it the key to the physics of the cosmos." In the sixteenth century, Johannes Kepler, in writing about the Golden, or "Divine Section," said that it described virtually all of creation and specifically symbolized God's creation of "like from like." Man is divided at the navel into Fibonacci proportions. The statistical average is approximately .618. The ratio holds true separately for men, and separately for women, a fine symbol of the creation of "like from like." Is all of mankind's progress also a creation of "like from like?"

The Golden Rectangle

The sides of a Golden Rectangle are in the proportion of 1.618 to 1. To construct a Golden Rectangle, start with a square of 2 units by 2 units and draw a line from the midpoint of one side of the square to one of the corners formed by the opposite side as shown in Figure 3-4.

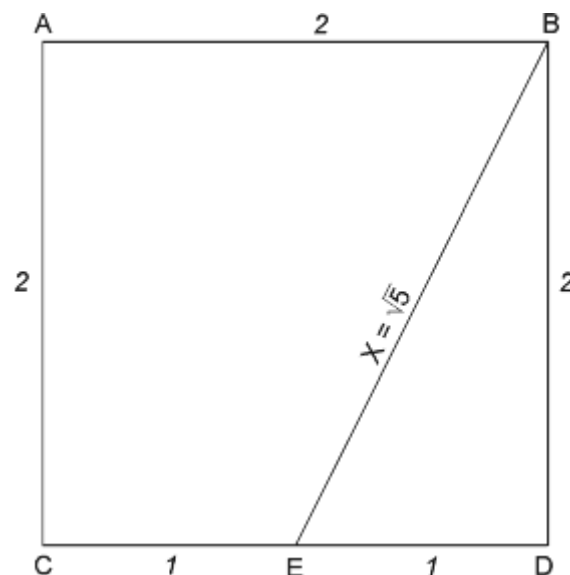


Figure 3-4

Triangle EDB is a right-angled triangle. Pythagoras, around 550 B.C., proved that the square of the hypotenuse (X) of a right-angled triangle equals the sum of the squares of the other two sides. In this case, therefore, $X^2 = 2^2 + 1^2$, or $X^2 = 5$. The length of the line EB , then, must be the square root of 5. The next step in the construction of a Golden Rectangle is to extend the line CD , making EG equal to the square root of 5, or 2.236, units in length, as shown in Figure 3-5. When completed, the sides of the rectangles are in the proportion of the Golden Ratio, so both the rectangle $AFGC$ and $BFGD$ are Golden Rectangles.

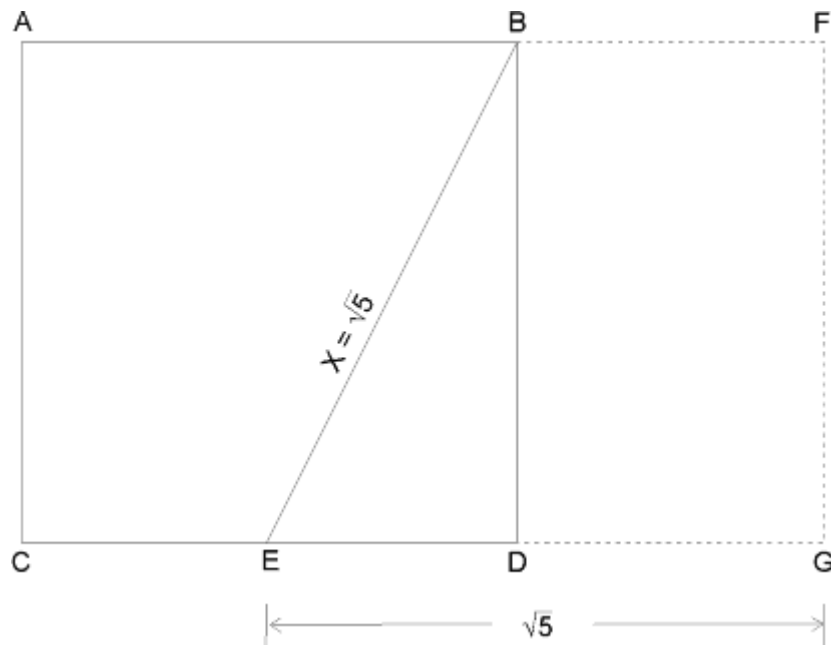


Figure 3-5

Since the sides of the rectangles are in the proportion of the Golden Ratio, then the rectangles are, by definition, Golden Rectangles.

Works of art have been greatly enhanced with knowledge of the Golden Rectangle. Fascination with its value and use was particularly strong in ancient Egypt and Greece and during the Renaissance, all high points of civilization. Leonardo da Vinci attributed great meaning to the Golden Ratio. He also found it pleasing in its proportions and said, "If a thing does not have the right look, it does not work." Many of his paintings had the right look because he used the Golden Section to enhance their appeal.

While it has been used consciously and deliberately by artists and architects for their own reasons, the *phi* proportion apparently does have an effect upon the viewer of forms. Experimenters have determined that people find the .618 proportion aesthetically pleasing. For instance, subjects have been asked to choose one rectangle from a group of different types of rectangles with the average choice generally found to be close to the Golden Rectangle shape. When asked to cross one bar with another in a way they liked best, subjects generally used one to divide the other into the *phi* proportion. Windows, picture frames, buildings, books and cemetery crosses often approximate Golden Rectangles.

As with the Golden Section, the value of the Golden Rectangle is hardly limited to beauty, but serves function as well. Among numerous examples, the most striking is that the double helix of DNA itself creates precise Golden Sections at regular intervals of its twists (see Figure 3-9).

While the Golden Section and the Golden Rectangle represent static forms of natural and man-made aesthetic beauty and function, the representation of an aesthetically pleasing dynamism, an orderly progression of growth or progress, can be made only by one of the most remarkable forms in the universe, the Golden Spiral.

8.6 The Golden Spiral

A Golden Rectangle can be used to construct a Golden Spiral. Any Golden Rectangle, as in Figure 3-5, can be divided into a square and a smaller Golden Rectangle, as shown in Figure 3-6. This process then theoretically can be continued to infinity. The resulting squares we have drawn, which appear to be whirling inward, are marked A, B, C, D, E, F and G.

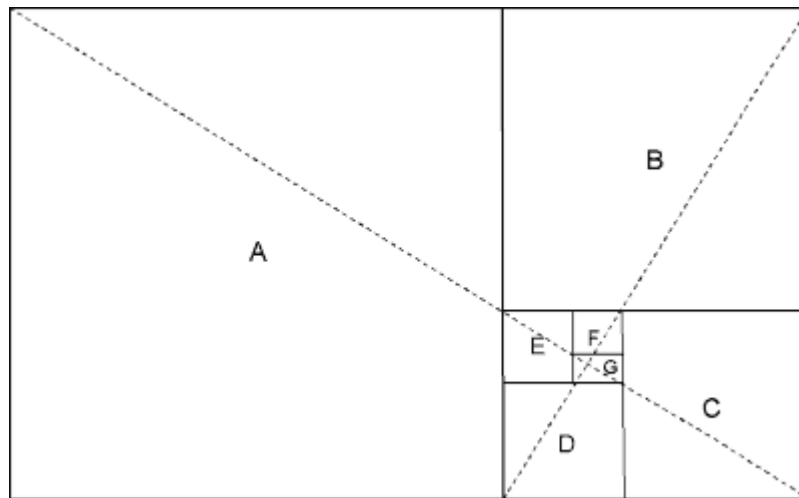


Figure 3-6

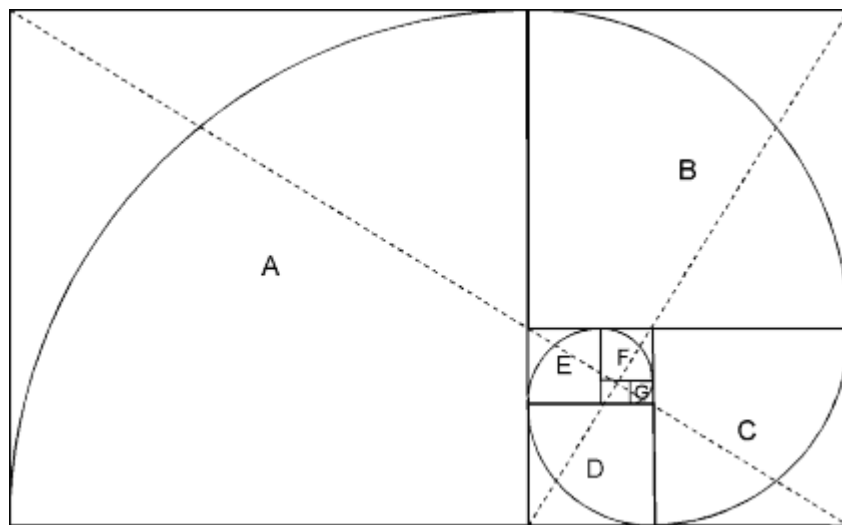
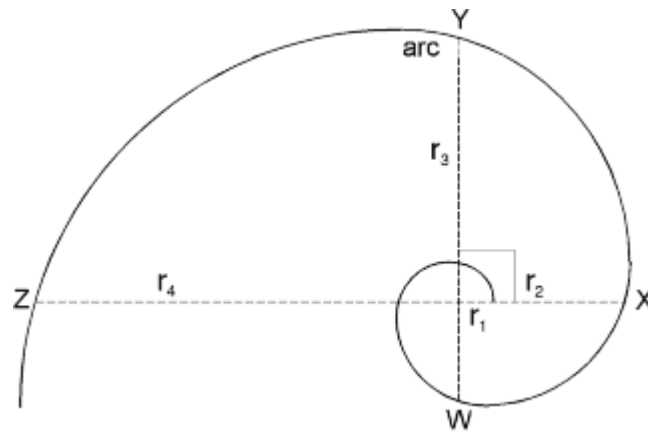


Figure 3-7

The dotted lines, which are themselves in golden proportion to each other, diagonally bisect the rectangles and pinpoint the theoretical center of the whirling squares. From near this central point, we can draw the spiral as shown in Figure 3-7 by connecting the points of intersection for each whirling square, in order of increasing size. As the squares whirl inward and outward, their connecting points trace out a Golden Spiral. The same process, but using a sequence of whirling triangles, also can be used to construct a Golden Spiral.

At any point in the evolution of the Golden Spiral, the ratio of the length of the arc to its diameter is 1.618. The diameter and radius, in turn, are related by 1.618 to the diameter and radius 90° away, as illustrated in Figure 3-8.



$$\frac{r_2}{r_1} = \frac{r_3}{r_2} = \frac{r_4}{r_3} = \dots = \frac{r_n}{r_{n-1}} = 1.618$$

$$\frac{d_2}{d_1} = \frac{d_3}{d_2} = \dots = \frac{d_n}{d_{n-1}} = 1.618$$

(where $d_1 = r_1 + r_3$, $d_2 = r_2 + r_4$, etc.)

$$\frac{\text{arcXY}}{\text{arcWX}} = \frac{\text{arcYZ}}{\text{arcXY}}, \text{ etc. } = \frac{\text{arcXZ}}{\text{arcWY}} = 1.618$$

$$\frac{\text{arcWY}}{\text{diam. (WY)}} = \frac{\text{arcXZ}}{\text{diam. (XZ)}}, \text{ etc. } = 1.618$$

Figure 3-8

8.7 The Golden Spiral

The Golden Spiral, which is a type of logarithmic or equiangular spiral, has no boundaries and is a constant shape. From any point on the spiral, one can travel infinitely in either the outward or inward direction. The center is never met, and the outward reach is unlimited. The core of a logarithmic spiral seen through a microscope would have the same look as its widest viewable reach from light years away. As David Bergamini, writing for *Mathematics* (in Time-Life Books' Science Library series) points out, the tail of a comet curves away from the sun in a logarithmic spiral. The epeira spider spins its web into a logarithmic spiral. Bacteria grow at an accelerating rate that can be plotted along a logarithmic spiral. Meteorites, when they rupture the surface of the Earth, cause depressions that correspond to a logarithmic spiral. Pine cones, sea horses, snail shells, mollusk shells, ocean waves, ferns, animal horns and the arrangement of seed curves on sunflowers and daisies all form logarithmic spirals. Hurricane clouds and the galaxies of outer space swirl in logarithmic spirals. Even the human finger, which is composed of three bones in Golden Section to one another, takes the spiral shape of the dying poinsettia leaf when curled. In Figure 3-9, we see a reflection of this cosmic influence in numerous forms. Eons of time and light years of space separate the pine cone and the spiraling galaxy, but the design is the same: a 1.618 ratio, perhaps the primary law governing dynamic natural phenomena. Thus, the Golden Spiral spreads before us in symbolic form as one of nature's grand designs, the image of life in endless expansion and contraction, a static law governing a dynamic process, the within and the without sustained by the 1.618 ratio, the Golden Mean.

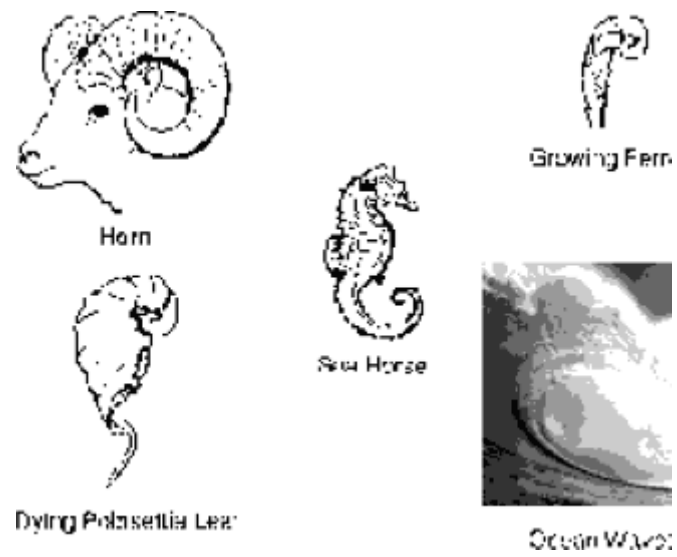


Figure 3-9a

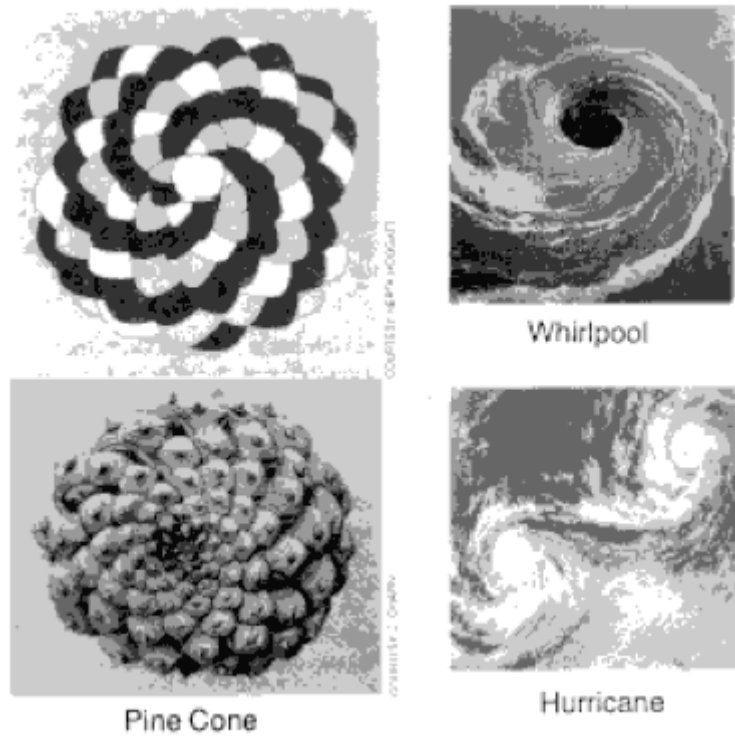


Figure 3-9b

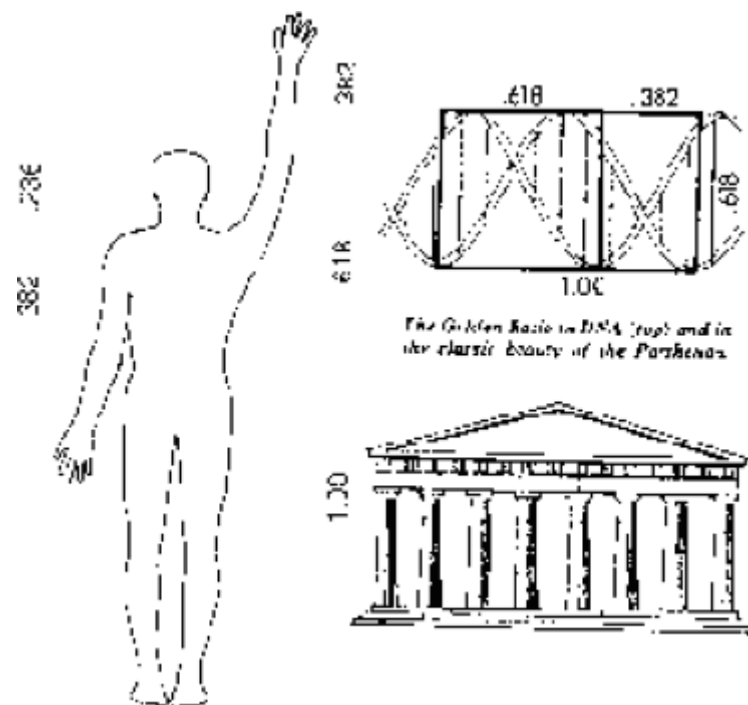


Figure 3-9c

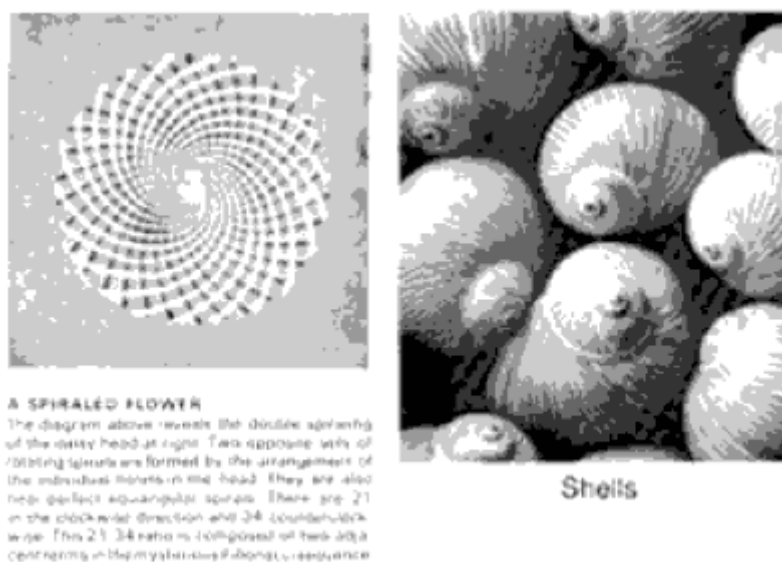
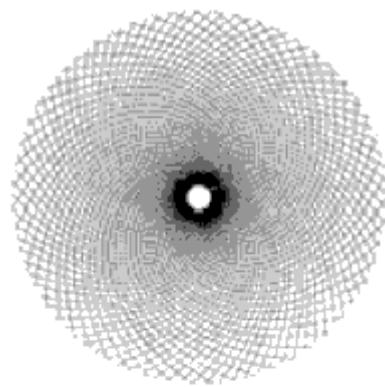
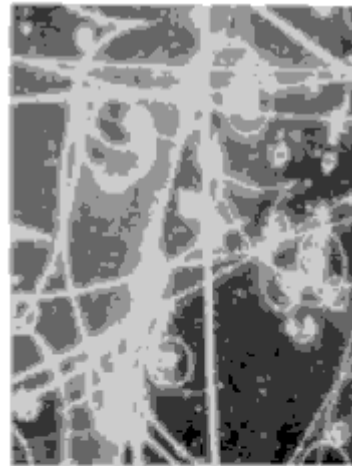


Figure 3-9d



Sunflower



Atomic Particles in
Bubble Chamber



Nautilus

Figure 3-9e

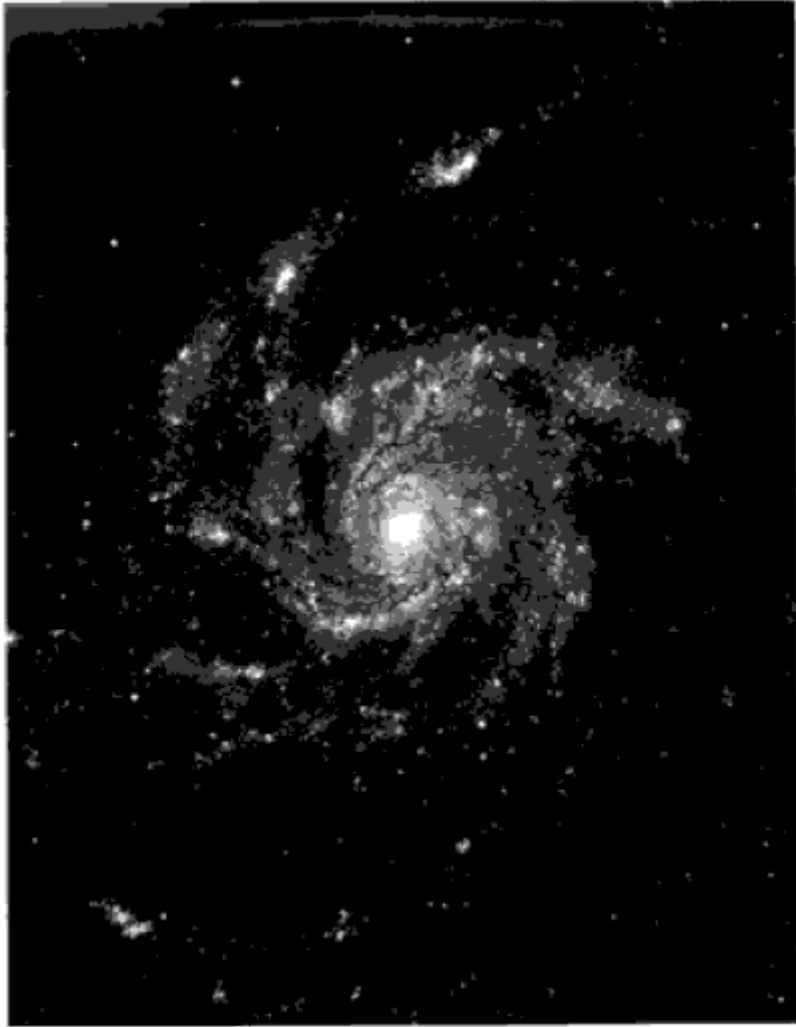


Figure 3-9f

Lesson 9

9.1 The Meaning of Phi

The value of this ubiquitous phenomenon was deeply understood and profoundly appreciated by the greatest intellects of the ages. History abounds with examples of exceptionally learned men who held a special fascination for this mathematical formulation. Pythagoras chose the five-pointed star, in which every segment is in golden ratio to the next smaller segment, as the symbol of his Order; celebrated 17th century mathematician Jacob Bernoulli had the Golden Spiral etched into his headstone; Isaac Newton had the same spiral carved on the headboard of his bed (owned today by the Gravity Foundation, New Boston, NH). The earliest known aficionados were the architects of the Gizeh pyramid in Egypt, who recorded the knowledge of *phi* in its construction nearly 5000 years ago. Egyptian engineers consciously incorporated the Golden Ratio in the Great Pyramid by giving its faces a slope height equal to 1.618 times half its base, so that the vertical height of the pyramid is at the same time the square root of 1.618 times half its base. According to Peter Tompkins, author of *Secrets of the Great Pyramid* (Harper & Row, 1971), "This relation shows Herodotus' report to be indeed correct, in that the square of the height of the pyramid is $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$, and the areas of the face $1 \times 1 = 1$." Furthermore, using these proportions, the Egyptian scientists (apparently in order to build a scale model of the Northern Hemisphere) used *pi* and *phi* in an approach so mathematically sophisticated that it accomplished the feat of squaring the circle and cubing the sphere (i.e., making them of equal area and volume), a feat which was not duplicated for well over four thousand years.

While the mere mention of the Great Pyramid may serve as an engraved invitation to skepticism (perhaps for good reason), keep in mind that its form reflects the same fascination held by pillars of Western scientific, mathematical, artistic and philosophic thought, including Plato, Pythagoras, Bernoulli, Kepler, DaVinci and Newton. Those who designed and built the pyramid were likewise demonstrably brilliant scientists, astronomers, mathematicians and engineers. Clearly they wanted to enshrine for millennia the Golden Ratio as something of transcendent importance. That such a caliber of people, who were later joined by some of the greatest minds of Greece and the Enlightenment in their fascination for this ratio, undertook this task is itself important. As for *why*, all we have is conjecture from a few authors. Yet that conjecture, however obtuse, curiously pertains to our own observations. It has been surmised that the Great Pyramid, for centuries after it was built, was used as a temple of initiation for those who proved themselves worthy of understanding the great universal secrets. Only those who could rise above the crude acceptance of things as they seemed to discover what, in actuality, they *were*, could be instructed in "the mysteries," i.e., the complex truths of eternal order and growth. Did such "mysteries" include *phi*? Tompkins explains, "The pharaonic Egyptians, says Schwaller de Lubicz, considered *phi* not as a number, but as a symbol of the creative function, or of reproduction in an endless series. To them it represented 'the fire of life, the male action of sperm, the *logos* [referenced in] the gospel of St. John.'" *Logos*, a Greek word, was defined variously by Heraclitus and subsequent pagan, Jewish and Christian philosophers as meaning the rational order of the universe, an immanent natural law, a life-giving force hidden within things, the universal structural force governing and permeating the world.

9.2 Conceptual Phi

Consider when reading such deep yet vague descriptions that these people could not clearly see what they sensed. They did not have graphs and the Wave Principle to make nature's growth pattern manifest and were doing the best they could to describe an organizational principle that they discerned as shaping the natural world. If these ancient philosophers were right that a universal structural force governs and permeates the world, should it not govern and permeate the world of man? If forms throughout the universe, including man's body, brain and DNA, reflect the form of *phi*, might man's activities reflect it as well? If *phi* is the life-force in the universe, might it be the impulse behind the progress in man's productive capacity? If *phi* is a symbol of the creative function, might it govern the creative activity of man? If man's progress is based upon production and reproduction "in an endless series," is it not reasonable that such progress has the spiraling form of *phi*, and that this form is discernible in the movement of the valuation of his productive capacity, i.e., the stock market? Just as the initiated Egyptians learned the hidden truths of order and growth in the universe behind the apparent randomness and chaos (something that modern "chaos theory" has finally rediscovered in the 1980s), so the stock market, in our opinion, can be understood properly if it is taken for what it

is rather than for what it crudely appears to be upon cursory consideration. The stock market is not a random, formless mess reacting to current news events but a remarkably precise recording of the formal structure of the progress of man.

Compare this concept with astronomer William Kingsland's words in *The Great Pyramid in Fact and in Theory* that Egyptian astronomy/astrology was a "profoundly esoteric science connected with the great cycles of man's evolution." The Wave Principle explains the great cycles of man's evolution and reveals how and why they unfold as they do. Moreover, it encompasses micro as well as macro scales, all of which are based upon a paradoxical principle of dynamism and variation within an unaltered form.

It is this form that gives structure and unity to the universe. Nothing in nature suggests that life is disorderly or formless. The word "universe" means "one order." If life has form, then we must not reject the probability that human progress, which is part of the reality of life, also has order and form. By extension, the stock market, which values man's productive enterprise, will have order and form as well. All technical approaches to understanding the stock market depend on the basic principle of order and form. Elliott's theory, however, goes beyond all others. It postulates that no matter how minute or how large the form, *the basic design remains constant*.

9.3 Phi and Elliott

Elliott, in his second monograph, used the title *Nature's Law — The Secret of the Universe* in preference to "The Wave Principle" and applied it to all sorts of human activity. Elliott may have gone too far in saying that the Wave Principle was *the* secret of the universe, as nature appears to have created numerous forms and processes, not just one simple design. Nevertheless, some of history's greatest scientists, mentioned earlier, would probably have agreed with Elliott's formulation. At minimum, it is credible to say that the Wave Principle is one of the most important secrets of the universe. Even this grandiose claim at first may appear to be only so much tall talk to practically-minded investors, and quite understandably so. The grand nature of the concept stretches the imagination and confounds the intellect, while its applicability is as yet unclear. First we must ask, can we both theorize and observe that there is indeed a principle that operates on the same mathematical basis in the heavens and earth as it does in the stock market?

The answer is yes. The stock market has the very same mathematical base as do these natural phenomena. The idealized Elliott concept of the progression of the stock market is an excellent base from which to construct the Golden Spiral, as Figure 3-10 illustrates with a rough approximation. In this construction, the top of each successive wave of higher degree is the touch point of the logarithmic expansion.

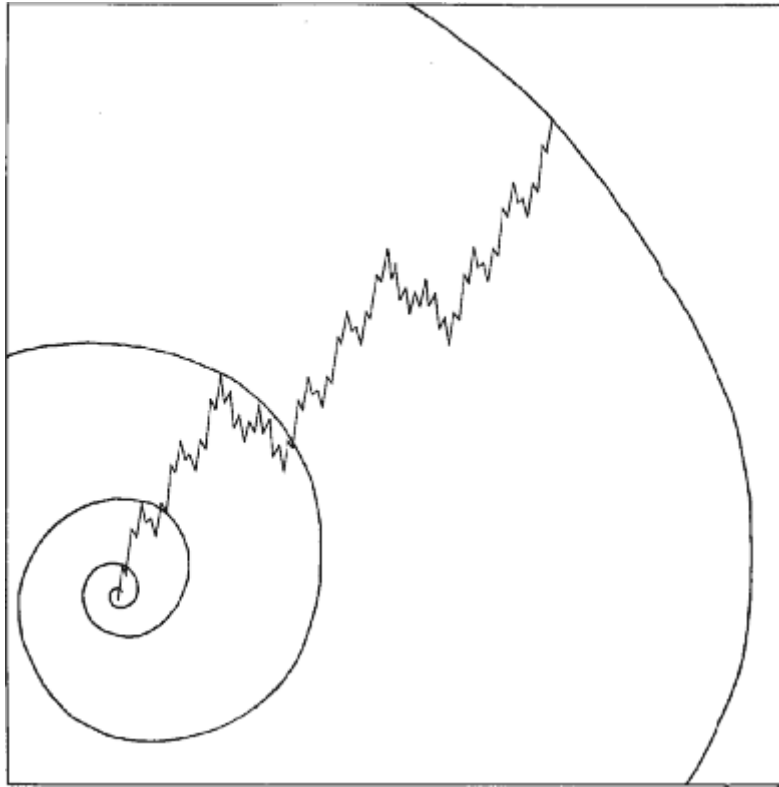


Figure 3-10

This result is possible because at every degree of stock market activity, a bull market subdivides into five waves and a bear market subdivides into three waves, giving us the 5-3 relationship that is the mathematical basis of the Elliott Wave Principle. We can generate the complete Fibonacci sequence, as we first did in Figure 1-4, by using Elliott's concept of the progression of the market. If we start with the simplest expression of the concept of a bear swing, we get one straight line decline. A bull swing, in its simplest form, is one straight line advance. A complete cycle is two lines. In the next degree of complexity, the corresponding numbers are 3, 5 and 8. As illustrated in Figure 3-11, this sequence can be taken to infinity.

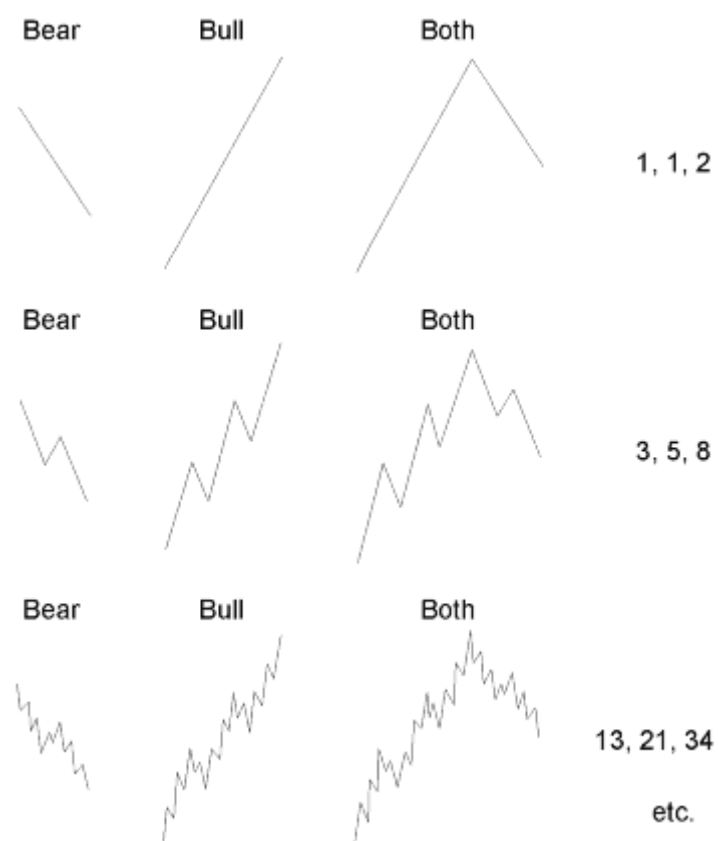


Figure 3-11

Lesson 10

10.1 Phi And The Stock Market

The stock market's patterns are repetitive (and fractal, to use today's terminology) in that the same basic pattern of movement that shows up in minor waves, using hourly plots, shows up in Supercycles and Grand Supercycles, using yearly plots. Figures 3-12 and 3-13 show two charts, one reflecting the hourly fluctuations in the Dow over a ten day period from June 25th to July 10th, 1962 and the other a yearly plot of the S&P 500 Index from 1932 to 1978 (courtesy of *The Media General Financial Weekly*). Both plots indicate similar patterns of movement despite a difference in the time span of over 1500 to 1. The long term formulation is still unfolding, as wave V from the 1974 low has not run its full course, but to date the pattern is along lines parallel to the hourly chart. Why? Because in the stock market, form is not a slave to the time element. Under Elliott's rules, both short and long term plots reflect a 5-3 relationship that can be aligned with the form that reflects the properties of the Fibonacci sequence of numbers. This truth suggests that collectively, man's emotions, in their expression, are keyed to this mathematical law of nature.

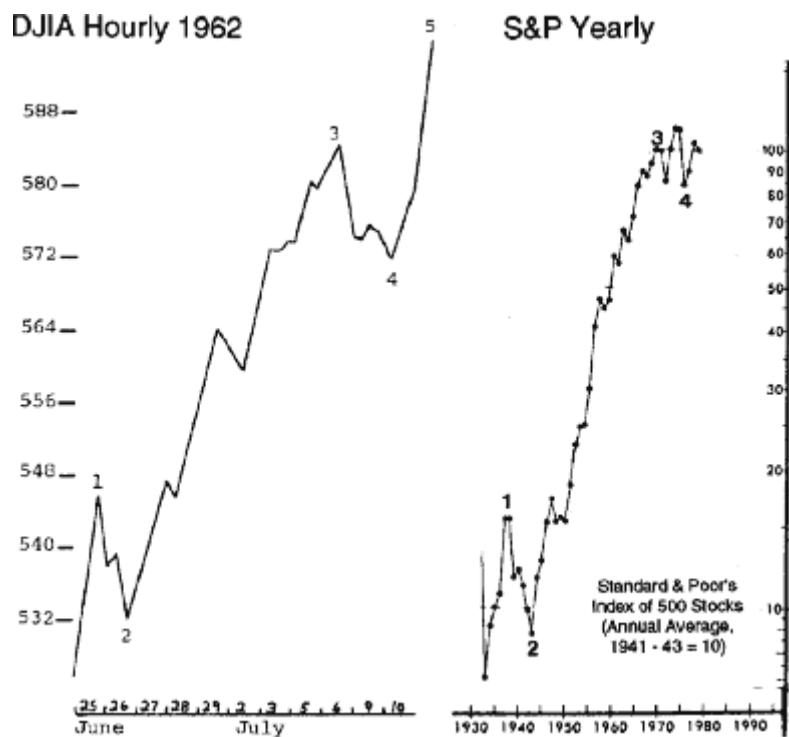


Figure 3-12

Figure 3-13

10.2 Phi And The Market

Now compare the formations shown in Figures 3-14 and 3-15. Each illustrates the natural law of the inwardly directed Golden Spiral and is governed by the Fibonacci ratio. Each wave relates to the previous wave by .618. In fact, the distances in terms of the Dow points themselves reflect Fibonacci mathematics. In Figure 3-14, showing the 1930-1942 sequence, the market swings cover approximately 260, 160, 100, 60, and 38 points respectively, closely resembling the declining list of Fibonacci ratios: 2.618, 1.618, 1.00, .618 and .382.

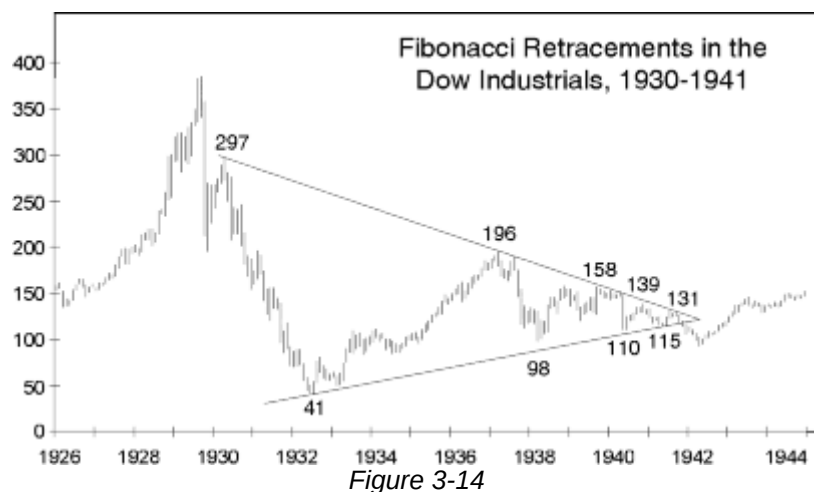


Figure 3-14

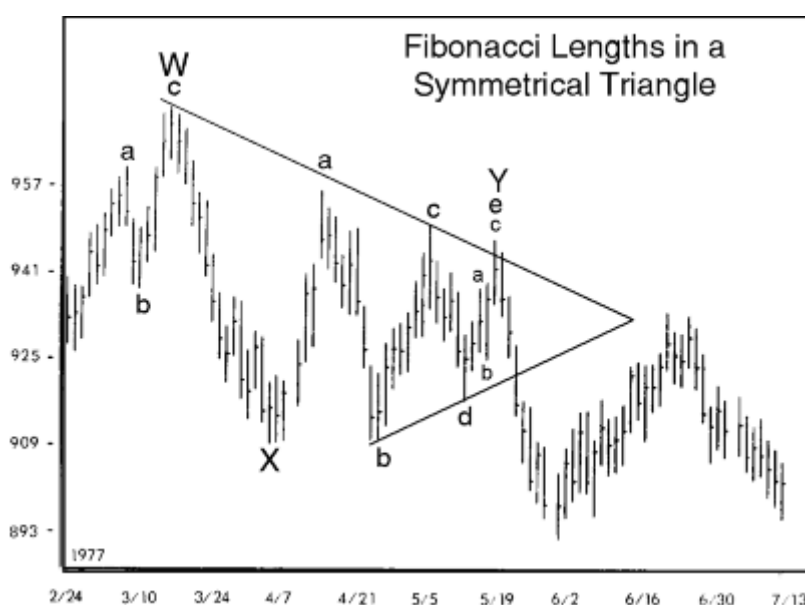


Figure 3-15

Starting with wave X in the 1977 upward correction shown in Figure 3-15, the swings are almost exactly 55 points (wave X), 34 points (waves A through C), 21 points (wave d), 13 points (wave a of e) and 8 points (wave b of e), the Fibonacci sequence itself. The total net gain from beginning to end is 13 points, and the apex of the triangle lies exactly on the level of the correction's beginning at 930, which is also the level of the peak of the subsequent reflex rally in June. Whether one takes the actual number of points in the waves as coincidence or part of the design, one can be certain that the precision manifest in the constant .618 ratio between each successive wave is not coincidence. Lessons 20 through 25 and 30 will elaborate substantially on the appearance of the Fibonacci ratio in market patterns.

10.3 Phi and Additive Growth

As we will show in subsequent lessons, the spiral-like form of market action is repeatedly shown to be governed by the Golden Ratio, and even Fibonacci numbers appear in market statistics more often than mere chance would allow. However, it is crucial to understand that while the numbers themselves do have theoretic weight in the grand concept of the Wave Principle, it is the *ratio* that is the fundamental key to growth patterns of this type. Although it is rarely pointed out in the

literature, the Fibonacci ratio results from this type of additive sequence no matter what two numbers start the sequence. The Fibonacci sequence is the basic additive sequence of its type since it begins with the number "1" (see Figure 3-17), which is the starting point of mathematical growth. However, we may also take any two *randomly selected numbers*, such as 17 and 352, and add them to produce a third, continuing in that manner to produce additional numbers. As this sequence progresses, the ratio between adjacent terms in the sequence always approaches the limit *phi* very quickly. This relationship becomes obvious by the time the eighth term is produced (see Figure 3-18). Thus, while the specific numbers making up the Fibonacci sequence reflect the ideal progression of waves in markets, the Fibonacci *ratio* is a fundamental law of geometric progression in which two preceding units are summed to create the next. That is why this ratio governs so many relationships in data series relating to natural phenomena of growth and decay, expansion and contraction, and advancement and retreat.

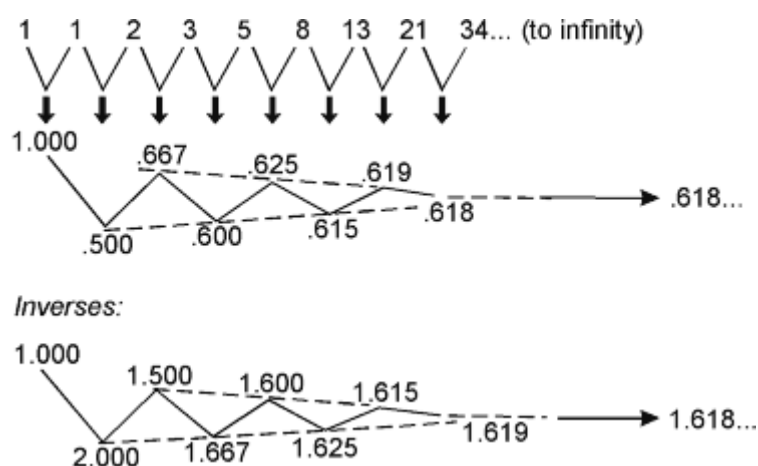


Figure 3-17

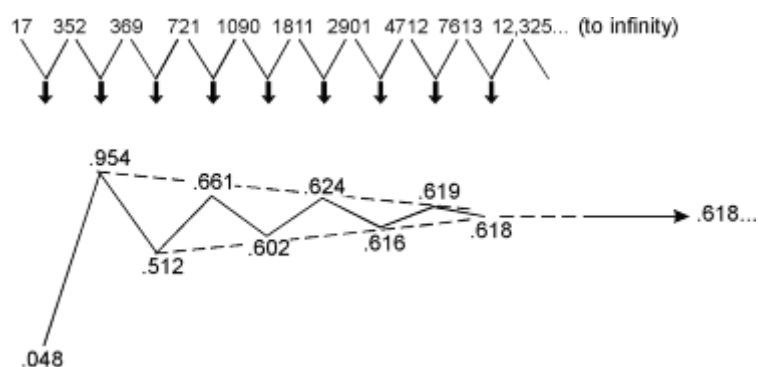


Figure 3-18

In its broadest sense, the Elliott Wave Principle proposes that the same law that shapes living creatures and galaxies is inherent in the spirit and activities of men *en masse*. The Elliott Wave Principle shows up clearly in the market because the stock market is the finest reflector of mass psychology in the world. It is a nearly perfect recording of man's social psychological states and trends, which produce the fluctuating valuation of his own productive enterprise, making manifest its very real patterns of progress and regress. What the Wave Principle says is that mankind's progress (of which the stock market is a popularly determined valuation) does not occur in a straight line, does not occur randomly, and does not occur cyclically. Rather, progress takes shape in a "three steps forward, two steps back" fashion, a form that nature prefers. In our opinion, the parallels between and Wave Principle and other natural phenomena are too great to be dismissed as just so much nonsense. On the balance of probabilities, we have come to the conclusion that there is a principle, everywhere present, giving shape to social affairs, and that Einstein knew what he was talking about when he said, "God does not play dice with the universe." The stock market is no exception, as mass

behavior is undeniably linked to a law that can be studied and defined. The briefest way to express this principle is a simple mathematical statement: the 1.618 ratio.

The *Desiderata*, by poet Max Ehrmann, reads, "You are a child of the Universe, no less than the trees and the stars; you have a right to be here. And whether or not it is clear to you, no doubt the Universe is unfolding as it should." Order in life? Yes. Order in the stock market? Apparently